Viet Nam National University, HCMC

University of Science

Institution of Information Technology

# Applied Statistics for Engineers and Scientists II

***Student:***
Tran My An (23125049)
Vo Tran Hien (23125056)
Phm Bao Kha (23125058)
Pham Minh Duc (23125080)

***Lecturer:***
Dr. Nguyen Thi Mong Ngoc
Dr. Nguyen Huu Toan

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# Member Contributions

This project is the result of the collaborative effort of all group members. The detailed breakdown of tasks and the contribution percentage for each member are outlined below:

**Pham Bao Kha (28%)**
- Led and contributed the majority of the content for Analysis 1.
- Researched and identified the dataset for Analysis 3.
- Participated in brainstorming and methodology discussions for Analysis 2.
- Developed the report template, structured the final document, and assisted with submission.

**Tran My An (28%)**
- Led and contributed the majority of the content for Analysis 2.
- Was the primary contributor to writing and finalizing the report file.
- Provided feedback and suggestions for Analysis 1 and Analysis 3.

**Vo Tran Hien (28%)**
- Performed data imputation and recovery for Analysis 1.
- Developed the baseline models for Analysis 1 and 2.
- Led and contributed the majority of the content for Analysis 3.

**Pham Minh Duc (16%)**
- Provided feedback and suggestions for Analysis 2 and 3.
- Assisted with tasks related to finalizing the report.

# 1 Activity 1 - CSM Dataset

## 1.1 Dataset Preview

The Conventional and Social Media (CSM) 2014 and 2015 dataset was obtained from the UC Irvine Machine Learning Repository. The original sources of the data include YouTube, Twitter, and IMDB. This dataset was provided by Mehreen Ahmed from the National University of Sciences and Technology, Islamabad, Pakistan, who also served as the data collector. The modules *Data Collector* and *Predictive Engine* were used to obtain the data. Originally, the dataset was designed for predictive analysis on the success of movies using machine learning algorithms.

There are only two unique values for *year* (2014 and 2015), 231 unique values for *movie name*, and 15 unique values for *genre*. The dataset primarily represents movies from IMDB, Wikipedia, YouTube, and Twitter, thus it tends to overrepresent American films while underrepresenting movies produced and released in other countries.

## 1.2 Data processing

### 1.2.1 Data Structure and Summary

We start by examining the structure and statistical summaries of each variable to gain an overview of the dataset's composition and identify potential issues. Key details include:

- The dataset comprises 231 observations across 14 variables.
- One character variable: Movie.

- 13 numeric variables: Year, Ratings, Genre, Gross, Budget, Screens, Sequel, Sentiment, Views, Likes, Dislikes, Comments, Aggregate.Followers.
- Genre and Sequel are numeric-coded but potentially interpretable as categorical.

**Insights from summaries:**

- Data spans years 2014-2015.
- Ratings range from 3.1 to 8.7 with a mean of 6.44.
- Gross revenue varies widely from $2,470$ to 643 million, indicating high variability.
- Missing values: Screens (10 missing, 4.33%), Aggregate.Followers (35 missing, 15.2%).
- Severe right-skewness in variables like Gross (mean 68 million vs. median 37 million), Budget, Views, and Likes.
- Evidenced by means much larger than medians and right-tailed histograms.

**Variable type: character**

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---|---|---|---|---|---|---|---|
| Movie | 0 | 1 | 3 | 41 | 0 | 231 | 0 |

**Variable type: numeric**

| skim_variable | n_missing | complete_rate | mean | sd | p0 | p25 | p50 | p75 | p100 | hist |
|---|---|---|---|---|---|---|---|---|---|---|
| Year | 0 | 1.00 | 2014.29 | 0.46 | 2014.0 | 2014.0 | 2014.0 | 2015.0 | 2015.0 | |
| Ratings | 0 | 1.00 | 6.44 | 0.99 | 3.1 | 5.8 | 6.5 | 7.1 | 8.7 | |
| Genre | 0 | 1.00 | 5.36 | 4.14 | 1.0 | 1.0 | 3.0 | 8.0 | 15.0 | |
| Gross | 0 | 1.00 | 68066033.20 | 88902891.22 | 2470.0 | 10300000.0 | 37400000.0 | 89350000.0 | 643000000.0 | |
| Budget | 1 | 1.00 | 47921730.05 | 54288246.07 | 70000.0 | 9000000.0 | 28000000.0 | 65000000.0 | 250000000.0 | |
| Screens | 10 | 0.96 | 2209.24 | 1463.77 | 2.0 | 449.0 | 2777.0 | 3372.0 | 4324.0 | |
| Sequel | 0 | 1.00 | 1.36 | 0.97 | 1.0 | 1.0 | 1.0 | 1.0 | 7.0 | |
| Sentiment | 0 | 1.00 | 2.81 | 7.00 | -38.0 | 0.0 | 0.0 | 5.5 | 29.0 | |
| Views | 0 | 1.00 | 3712851.29 | 4511104.24 | 698.0 | 623302.0 | 2409338.0 | 5217379.5 | 32626778.0 | |
| Likes | 0 | 1.00 | 12732.54 | 28825.48 | 1.0 | 1776.5 | 6096.0 | 15247.5 | 370552.0 | |
| Dislikes | 0 | 1.00 | 679.05 | 1243.93 | 0.0 | 105.5 | 341.0 | 697.5 | 13960.0 | |
| Comments | 0 | 1.00 | 1825.70 | 3571.04 | 0.0 | 248.5 | 837.0 | 2137.0 | 38363.0 | |
| Aggregate.Followers | 35 | 0.85 | 3038193.45 | 4886278.07 | 1066.0 | 183025.0 | 1052600.0 | 3694500.0 | 31030000.0 | |

Figure 1: Data summary

This skewness suggests non-normality and potential heteroscedasticity, which could violate linear regression assumptions. It may necessitate transformations like logarithmic or square-root adjustments during modeling.

### 1.2.2 Missing Data Analysis

A thorough analysis of missing data is required to choose an appropriate handling strategy. Improper handling could lead to biased estimates or loss of information. We first visualize the extent and pattern of missingness to identify variables and combinations affected.

**Key findings:**

- Aggregate.Followers: highest missing at 15.2% (35 cases).
- Screens: 4.3% (10 cases).
- Budget: 0.433% (1 case).
- Other variables: no missing values.

**The vis_miss plot reveals:**

- Missingness is sporadic.
- Primarily isolated to specific columns.
- No widespread row-wise patterns.

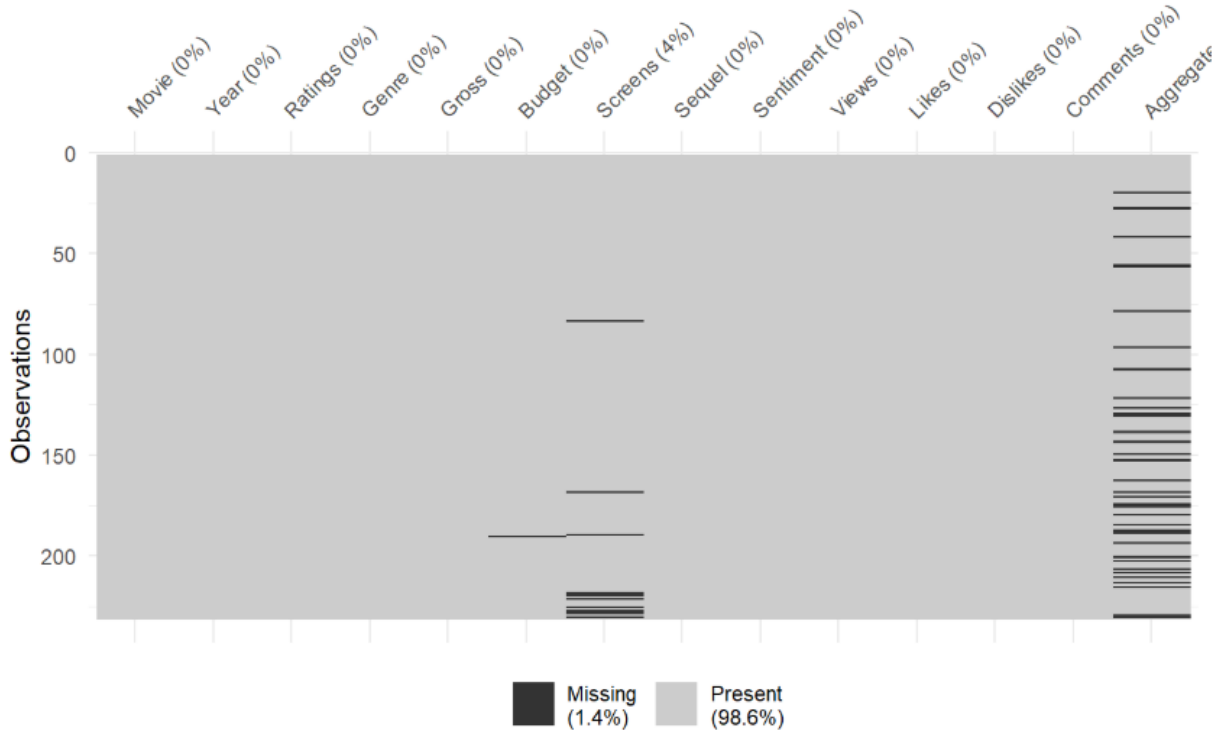



Figure 2: The vis_miss plot provides a heatmap-like overview of missing data locations.

**The aggr plot confirms:**

- Most missingness in Aggregate.Followers alone.
- Minimal overlap across variables.
- Suggests missing data not systematically affecting entire observations but specific features.

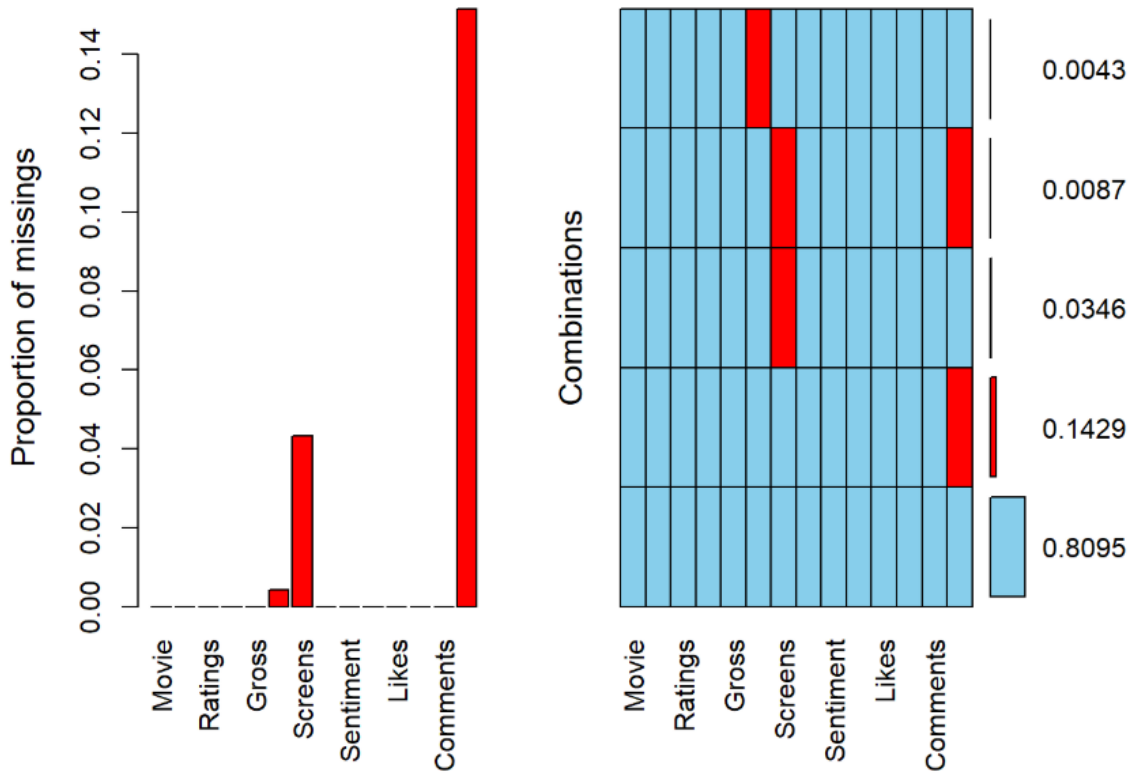



Figure 3: The aggr plot shows the amount of missingness.

### 1.2.3 Investigating the Missing Data Mechanism

We test the missing data mechanism to inform imputation choices. Possible mechanisms:

- Missing Completely at Random (MCAR): missingness unrelated to any data (allows simple imputation).
- Missing at Random (MAR): missingness depends on observed variables (requires model-based imputation).

**Methods used:**

- Logistic regression: predicts missingness indicator for Aggregate.Followers using Views, Likes, Budget, Screens, Ratings.

**Significant effects:**

- Budget (estimate -1.568e-08, p=0.031): negative, lower-budget movies more likely to miss follower data (possibly less social media tracking).
- Ratings (estimate 0.574, p=0.016): positive, highly rated niche films may lack follower metrics.

**Model's deviance reduction supports MAR. T-tests reinforce:**

- Budget means: missing 24.5$M$ vs. non-missing 52.1$M$ (p<0.001).
- Likes: marginal difference (p=0.057).

**Little's MCAR test:**

- Statistic=45.2, df=31, p=0.048: rejects MCAR, confirming systematic missingness.

```
## # A tibble: 1 × 4
##   statistic    df p.value missing.patterns
##       <dbl> <dbl>   <dbl>            <int>
## 1      45.2    31  0.0482                5
```

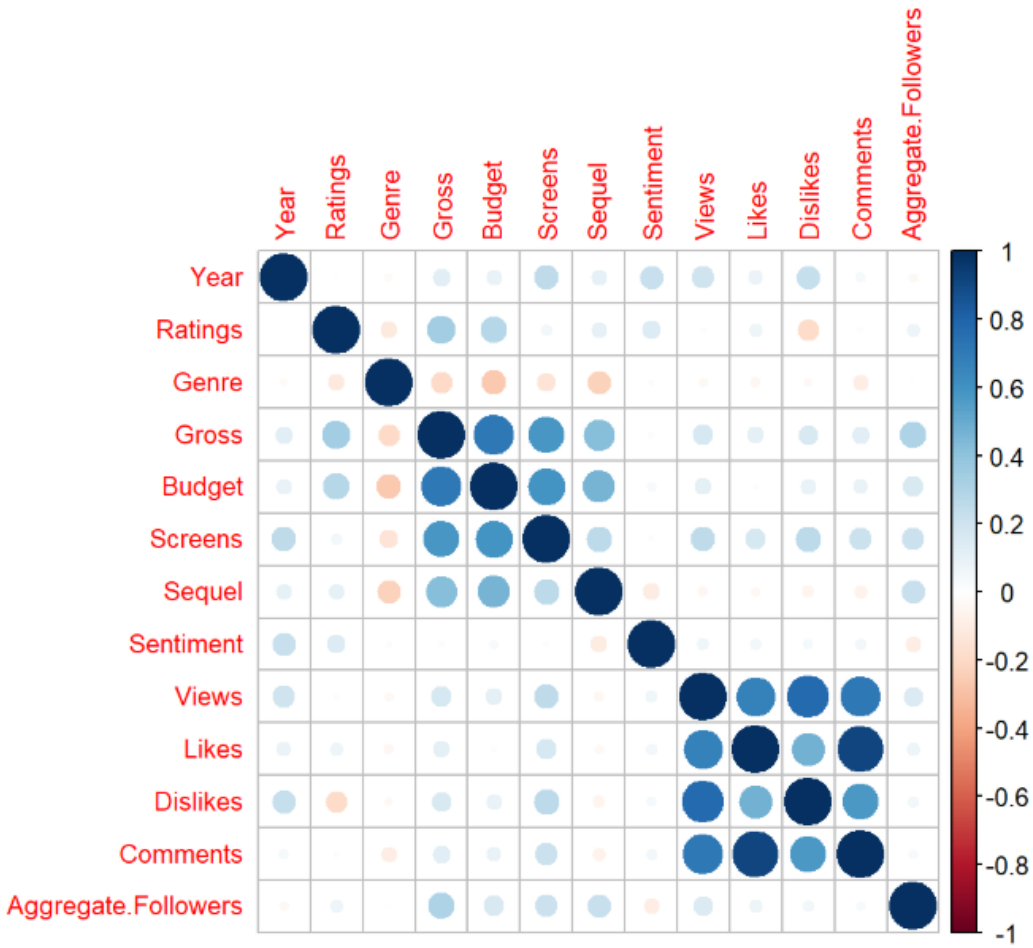



Figure 4: Missing correlations.

**Correlation matrix (pairwise complete observations):**

- Shows moderate relationships (e.g., Budget-Gross  0.6).
- Can be leveraged for imputation under MAR.

### 1.2.4 Data Imputation

Given strong evidence for MAR, we employ **missForest**:

- A non-parametric random forest-based method.
- Handles mixed variable types.
- Captures complex interactions effectively.

**Process:**

- Iteratively imputes missing values by training random forests on observed data.
- Uses 100 trees for robustness.
- Three iterations achieve convergence.

**Results:**

- Low out-of-bag (OOB) errors: NRMSE=0.248 (numeric), PFC=0 (categorical).
- Indicates high imputation accuracy.

**Post-imputation checks:**

- No remaining missing values.
- No negative imputations in non-negative variables (Budget, Gross, Aggregate.Followers).
- Preserves data integrity and realism.

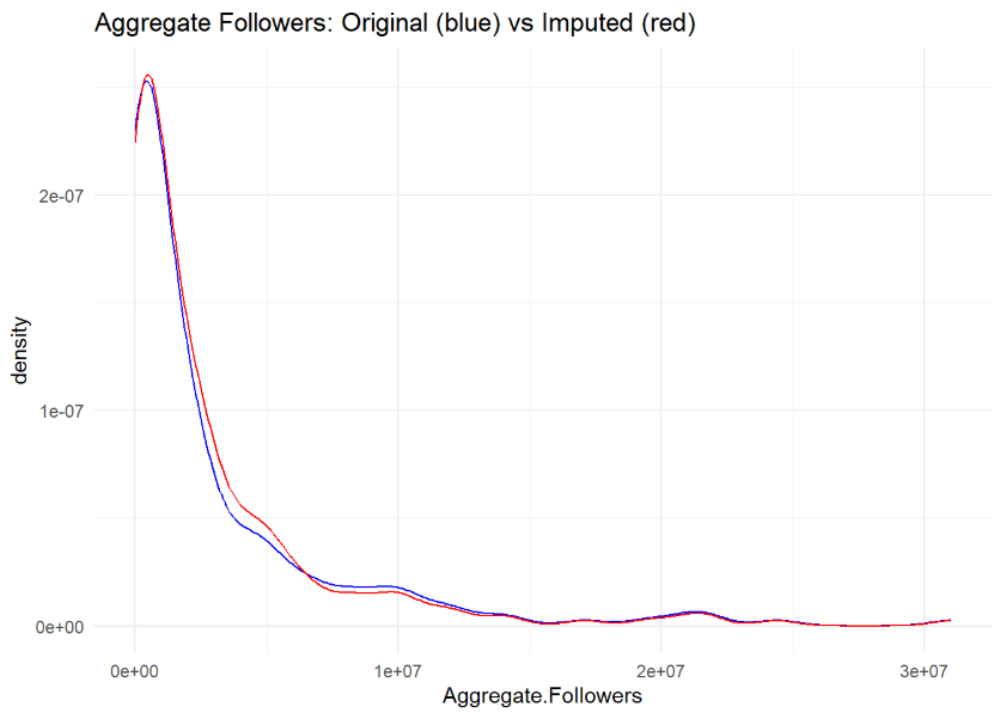
### 1.2.5 Post-Imputation Exploratory Data Analysis (EDA)

Before modeling, we perform final EDA on the complete imputed dataset. **Goals:**

- Verify imputation quality.
- Understand relationships.
- Confirm data characteristics for model readiness.

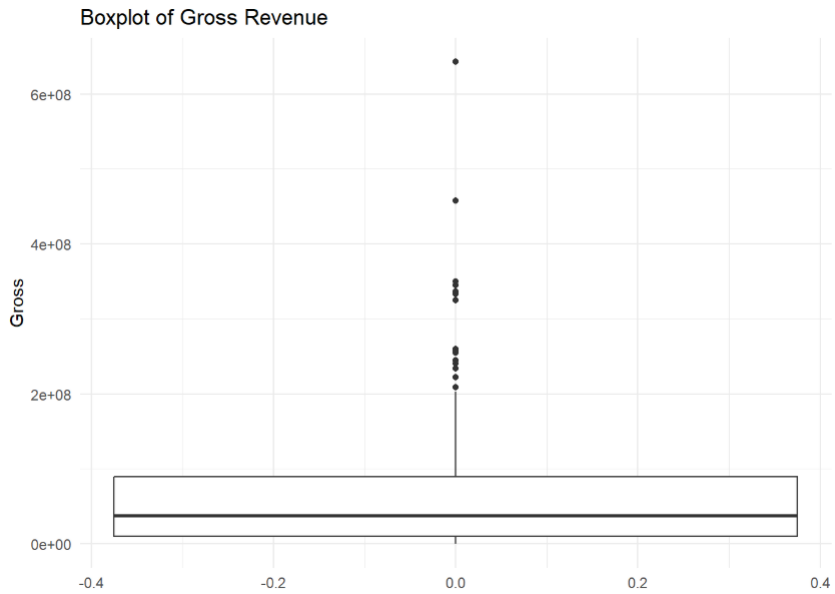
**Key visualizations and insights:**

- Near-perfect overlap between original (blue) and imputed (red) distributions.
- Preserves shape, scale, and multimodality.
- No introduced artifacts.



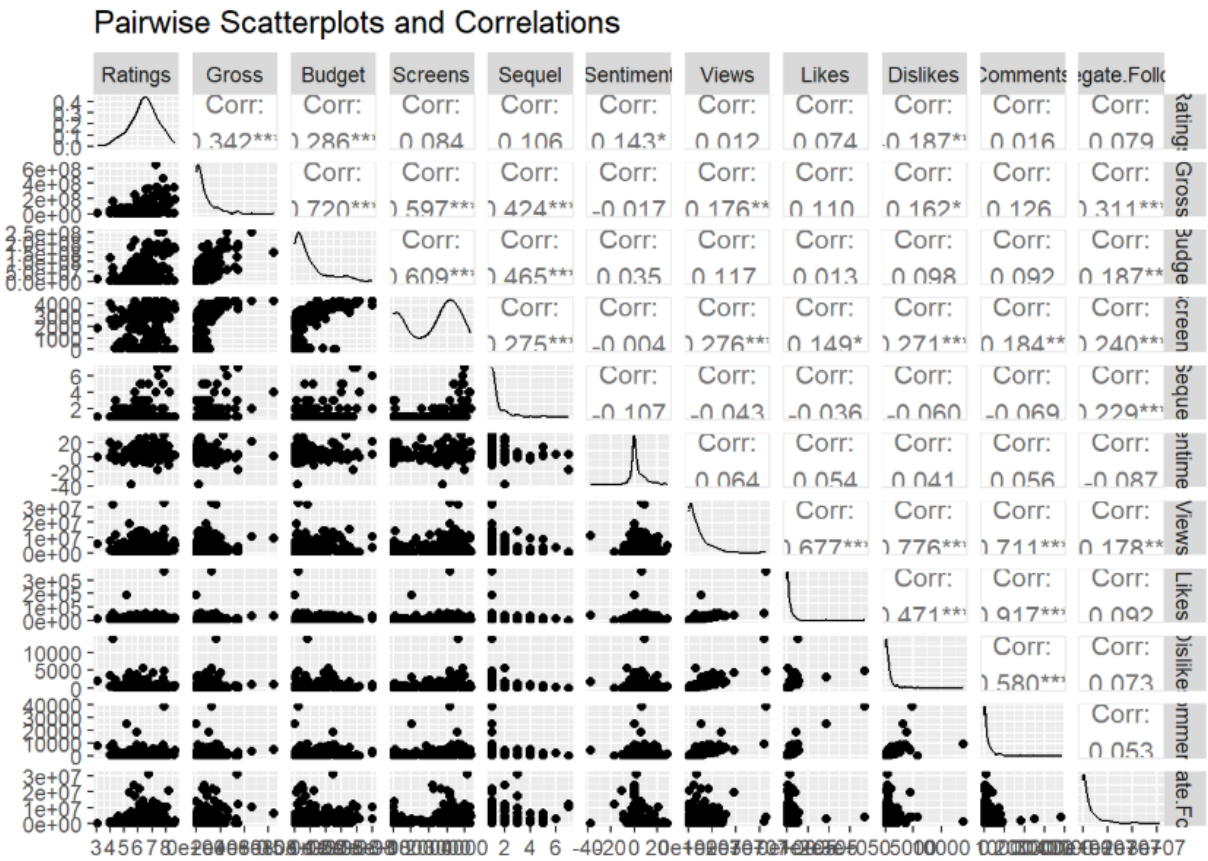
**Boxplot for Gross:**

- Highlights severe right-skewness.
- Long upper tail.
- Numerous outliers above 200 million.
- Underscores need for transformations (e.g., square-root or log) to normalize and stabilize variance.




**Pairwise scatterplots and correlations:**

- Strong positive associations: Budget-Gross $r \approx 0.7$, Screens-Gross $r \approx 0.6$.
- Potential multicollinearity: Views-Likes r>0.8.
- Non-linear patterns: e.g., quadratic in Ratings.
- Guides feature selection and transformations in modeling.




## 1.3 Dataset Splitting

To properly assess the model's predictive performance on unseen data, the complete, imputed dataset was partitioned. A standard 80/20 split was employed, allocating 80% of the data (184 observations) to the training set for model development and the remaining 20% (47 observations) to the validation set for final evaluation. This partitioning was performed randomly, with a fixed seed to ensure reproducibility.

## 1.4 Model selection

This section details the iterative process of selecting the best linear regression model to predict `Gross` revenue. The methodology involved constructing an initial comprehensive model, followed by a systematic process of feature selection, assumption diagnostics, outlier handling, and variable transformations to arrive at a final, robust model.

### 1.4.1 Initial Full Model

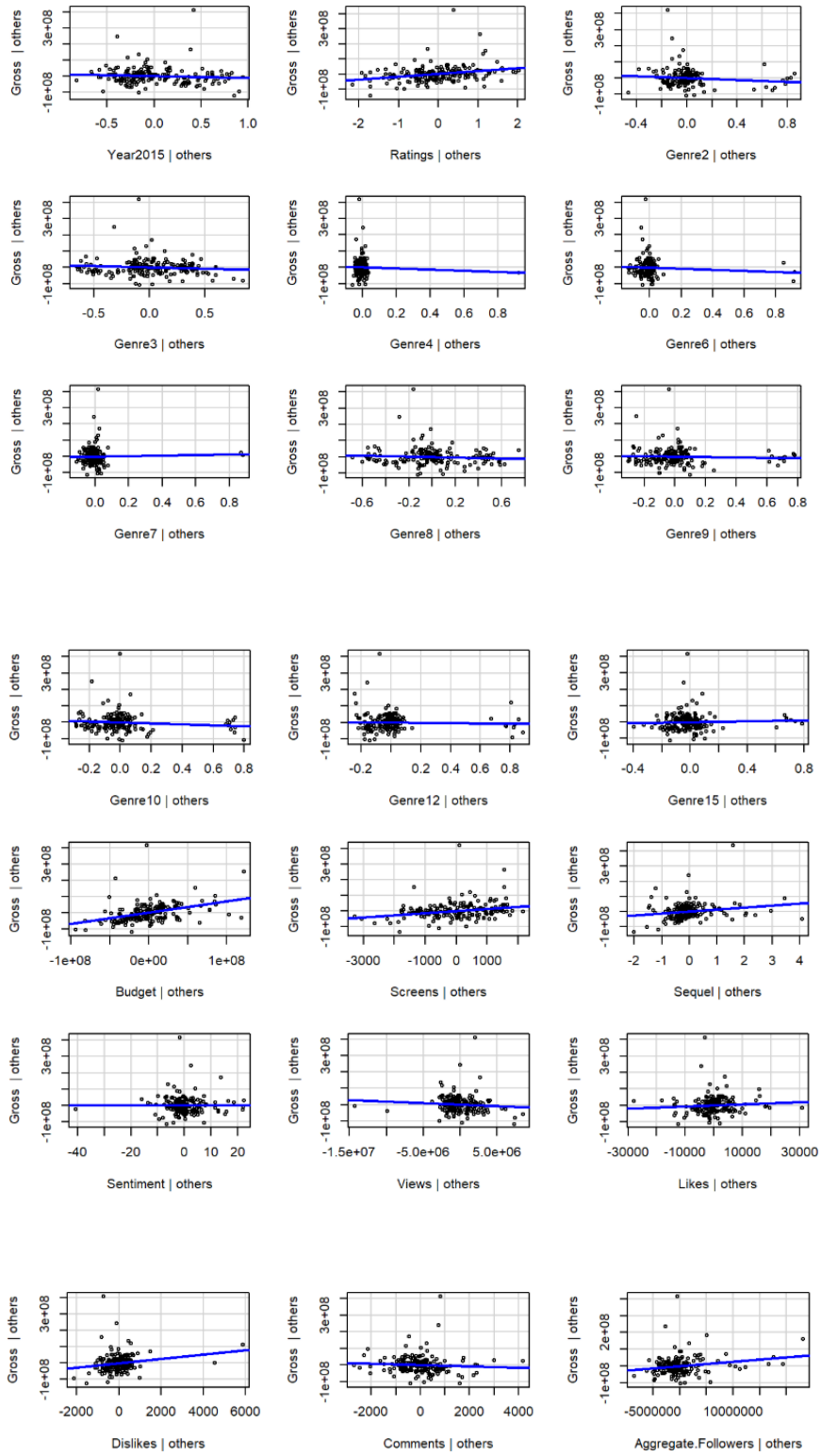
As a baseline, a multiple linear regression model was fitted using all available predictors in the training dataset. This "full model" serves as a starting point for subsequent refinement.

```
##
## Call:
## lm(formula = Gross ~ ., data = train)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -112259452  -30324174   -4394688   19041975  416247723
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -1.310e+08  3.657e+07  -3.582 0.000450 ***
## Year2015            -1.322e+07  1.132e+07  -1.168 0.244569
## Ratings              1.907e+07  5.308e+06   3.592 0.000435 ***
## Genre2              -3.099e+07  2.129e+07  -1.455 0.147556
## Genre3              -1.662e+07  1.393e+07  -1.193 0.234611
## Genre4              -3.419e+07  5.956e+07  -0.574 0.566714
## Genre6              -3.338e+07  3.496e+07  -0.955 0.341097
## Genre7               1.835e+07  4.322e+07   0.425 0.671699
## Genre8              -1.439e+07  1.447e+07  -0.994 0.321479
## Genre9              -1.478e+07  2.077e+07  -0.711 0.477802
## Genre10             -3.188e+07  2.125e+07  -1.500 0.135519
## Genre12             -1.001e+07  2.126e+07  -0.471 0.638530
## Genre15              1.258e+07  2.432e+07   0.517 0.605760
## Budget               6.944e-01  1.245e-01   5.577    1e-07 ***
## Screens              1.396e+04  4.090e+03   3.413 0.000812 ***
## Sequel               1.360e+07  5.454e+06   2.494 0.013638 *
## Sentiment           -4.649e+03  6.332e+05  -0.007 0.994151
## Views               -1.855e+00  1.803e+00  -1.029 0.304892
## Likes                6.158e+02  6.468e+02   0.952 0.342457
## Dislikes             1.304e+04  5.774e+03   2.258 0.025296 *
## Comments            -4.026e+03  4.699e+03  -0.857 0.392848
## Aggregate.Followers  2.535e+00  1.020e+00   2.486 0.013941 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 57220000 on 162 degrees of freedom
## Multiple R-squared:  0.6513, Adjusted R-squared:  0.6061
## F-statistic: 14.41 on 21 and 162 DF,  p-value: < 2.2e-16
```

Figure 5: Summary of the initial full linear regression model.

**Interpretation:**

- **Overall Fit:** The model is statistically significant ($F$-statistic $p$-value $< 0.001$) and explains about 61% of the variance in *Gross* ($R^2 = 0.6061$).
- **Key Predictors:** *Budget*, *Screens*, *Ratings*, and *Aggregate.Followers* are highly significant predictors.




Conclusion: The model is overly complex, with many variables (e.g., `Sentiment`, most `Genre` categories) showing no statistical significance. This indicates a clear need for feature selection to create a more parsimonious and interpretable model.

### 1.4.2 Feature Selection

To simplify the model and retain only the most impactful predictors, a dual-pronged approach of stepwise and best subset selection was employed.

### Backward Stepwise Selection

This process began with the full model and systematically eliminated the least significant variable at each iteration based on two criteria: the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC).

#### 1.4.2.1 Interpretation:

- **AIC Model:** The selection process using AIC resulted in a model retaining 6 predictor variables: `Ratings`, `Budget`, `Screens`, `Sequel`, `Dislikes`, and `Aggregate.Followers`.
- **BIC Model:** The BIC, which applies a stricter penalty for model complexity, selected a more parsimonious model with only 5 predictors, excluding `Dislikes`.

All variables in the BIC-selected model were highly significant. These 6 variables identified by either AIC or BIC were considered the strongest candidates for the next stage of analysis.

### Best Subset Selection

This method exhaustively evaluated all possible model combinations from the 6 candidate variables, assessing each model size against multiple statistical criteria.

Table 1: Comparison of Model Selection Criteria from Best Subsets.

```
##   (Intercept) Ratings Budget Screens Sequel Dislikes Aggregate.Followers
## 1        TRUE   FALSE   TRUE   FALSE  FALSE    FALSE               FALSE
## 2        TRUE   FALSE   TRUE    TRUE  FALSE    FALSE               FALSE
## 3        TRUE   FALSE   TRUE    TRUE  FALSE    FALSE                TRUE
## 4        TRUE    TRUE   TRUE    TRUE  FALSE    FALSE                TRUE
## 5        TRUE    TRUE   TRUE    TRUE   TRUE    FALSE                TRUE
## 6        TRUE    TRUE   TRUE    TRUE   TRUE     TRUE                TRUE
```

```
##   Num_Vars R_Squared Adj_R_Squared Mallows_Cp       BIC
## 1        1 0.5347906     0.5322345  44.636185 -130.3794
## 2        2 0.5707160     0.5659726  29.288845 -139.9523
## 3        3 0.5968354     0.5901160  18.676564 -146.2878
## 4        4 0.6147660     0.6061574  12.018407 -149.4437
## 5        5 0.6257568     0.6152443   8.711262 -149.5547
## 6        6 0.6334426     0.6210169   7.000000 -148.1579
```

#### 1.4.2.2 Interpretation:

The table evaluates the trade-off between model fit and complexity.

- **Adjusted R-Squared:** This metric, which penalizes the inclusion of non-informative predictors, peaks at 0.621 for the 6-variable model.
- **Mallows' Cp:** The ideal model has a Cp value that is both small and close to the number of parameters (p = variables + 1). The 6-variable model is optimal by this measure, with a Cp of 7.0, which is exactly p.

- **BIC:** This criterion, which most heavily penalizes complexity, reaches its minimum (best) value of -149.55 for the 5-variable model.

#### 1.4.2.3 Conclusion:

Given that two of the three key metrics (Adjusted $R^2$ and Mallows' Cp) favor the 6-variable model, it was selected as the best choice at this stage, offering a strong balance between explanatory power and parsimony.

### 1.4.3 Diagnostics of the Selected Model

The selected 6-variable model was then subjected to diagnostic checks to ensure it met the core assumptions of linear regression.

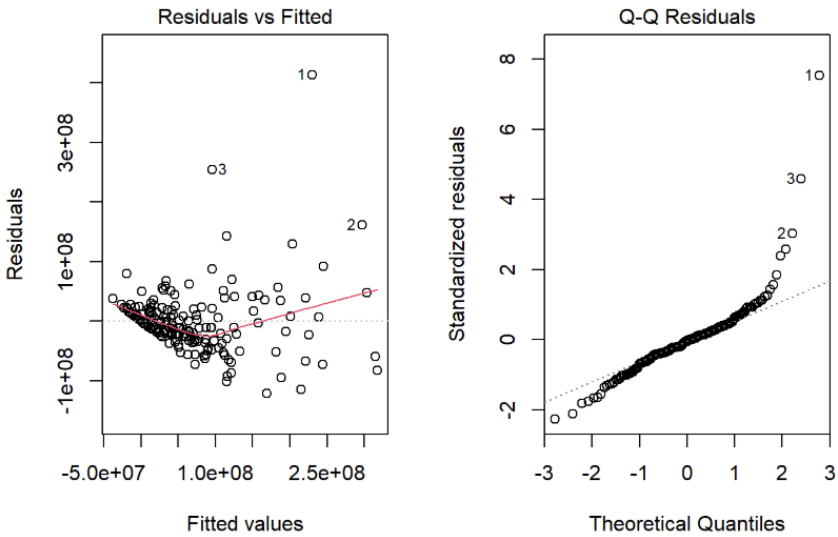



Figure 6: Diagnostic plots for the 6-variable model.

#### 1.4.3.1 Interpretation:

The diagnostic plots and statistical tests revealed clear violations of regression assumptions.

- **Linearity & Homoscedasticity:** The Residuals vs. Fitted plot displayed a distinct funnel shape, indicating that the variance of the residuals increases with the fitted values. This is a clear sign of heteroscedasticity.
- **Normality:** The Normal Q-Q plot showed that the residuals deviated heavily from the theoretical line, particularly at the tails. The Shapiro-Wilk test confirmed this, rejecting the null hypothesis of normality ($p < 0.001$).
- **Homoscedasticity Test:** The Breusch-Pagan test yielded a significant p-value ($< 0.05$), formally confirming the presence of heteroscedasticity.
- **Multicollinearity:** All Variance Inflation Factor (VIF) values were low ($<2.1$), indicating that multicollinearity was not an issue.

These violations necessitated further model refinement, starting with an investigation of influential outliers.

### 1.4.4 Outlier Detection and Removal

Influential data points can disproportionately skew model coefficients and diagnostic results. An influence plot was generated to identify such points based on their Cook's distance, leverage (hat values), and studentized residuals.

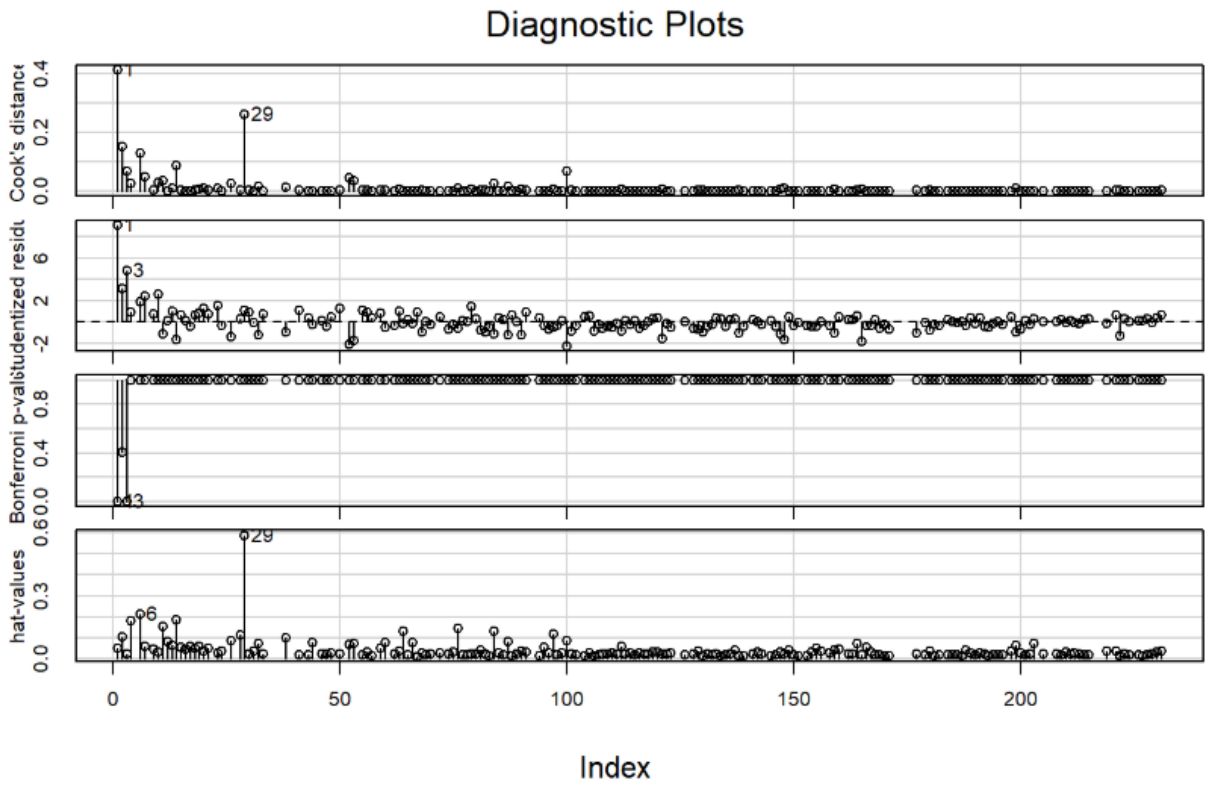



Figure 7: Influence Plot for Outlier Detection.

#### 1.4.4.1 Interpretation:

The influence plot identified observations 1 and 29 as highly influential. Both points exhibited high Cook's distances, and point 29 also showed high leverage, suggesting they had a substantial impact on the model's parameters.

### 1.4.5 Test for Outlier Removal

To confirm their influence, points 1 and 29 were removed from the training data, and the 6-variable model was refit.

```
##
## Call:
## lm(formula = Gross ~ Ratings + Budget + Screens + Sequel + Dislikes +
##     Aggregate.Followers, data = train_cleaned)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -124992842  -23362947   -1493183   16553533  257012206
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -1.024e+08  2.530e+07  -4.048 7.75e-05 ***
## Ratings              1.359e+07  3.761e+06   3.612 0.000396 ***
## Budget               7.541e-01  8.774e-02   8.595 4.52e-15 ***
## Screens              1.279e+04  3.041e+03   4.206 4.15e-05 ***
## Sequel               6.136e+06  4.178e+06   1.469 0.143701
## Dislikes             3.611e+02  4.322e+03   0.084 0.933504
## Aggregate.Followers  2.725e+00  7.880e-01   3.458 0.000683 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 46120000 on 175 degrees of freedom
## Multiple R-squared:  0.6839, Adjusted R-squared:  0.6731
## F-statistic: 63.12 on 6 and 175 DF,  p-value: < 2.2e-16
```

#### 1.4.5.1 Interpretation:

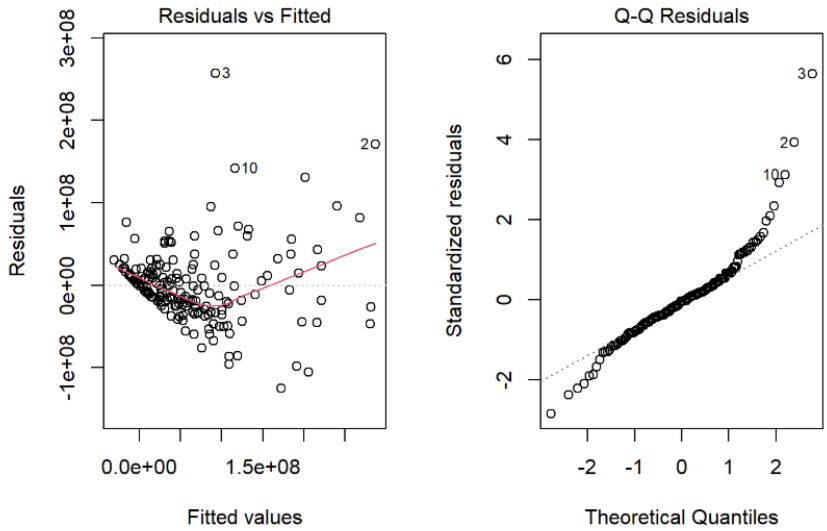
The removal of these two points resulted in a marked improvement in model performance. The Adjusted R-squared increased significantly from 0.621 to 0.673, and the Residual Standard Error decreased from $56.12 million to $46.12 million. This substantial enhancement confirmed that the points were indeed influential outliers, and their removal was justified for building a more robust model.

### 1.4.6 Re-selection After Outlier Removal

With the influential outliers removed, the backward selection process using AIC was repeated on the cleaned dataset to determine the optimal set of predictors.

#### 1.4.6.1 Interpretation:

After cleaning the data, the AIC-based selection process removed the `Dislikes` variable, as it was no longer statistically significant. This resulted in a more parsimonious 5-variable model. However, subsequent diagnostic checks revealed that the issues of heteroscedasticity and non-normality of residuals persisted, pointing to the need for variable transformations to fully satisfy the linear regression assumptions.




### 1.4.7 Variable Transformations

To address the remaining assumption violations stemming from data skewness, transformations were applied to both the predictor variables and the response variable, `Gross`.

### 1.4.8 Predictor Transformations: Power Transform for Non-Categorical Variables

The `powerTransform` function was used on the cleaned training data to find the optimal Box-Cox power transformations ($\lambda$) for the non-categorical predictors to improve symmetry and linearity.

```
## bcPower Transformations to Multinormality
##                      Est Power Rounded Pwr Wald Lwr Bnd Wald Upr Bnd
## Ratings                 1.8819        2.00       1.1207       2.6432
## Budget                  0.2137        0.21       0.1340       0.2933
## Screens                 0.6341        0.63       0.5135       0.7547
## Sequel                 -4.6427       -4.64      -5.4525      -3.8329
## Aggregate.Followers     0.1910        0.19       0.1265       0.2554
##
## Likelihood ratio test that transformation parameters are equal to 0
##  (all log transformations)
##                                    LRT df       pval
## LR test, lambda = (0 0 0 0 0) 494.9887  5 < 2.22e-16
##
## Likelihood ratio test that no transformations are needed
##                                    LRT df       pval
## LR test, lambda = (1 1 1 1 1) 1163.982  5 < 2.22e-16
```

Figure 8: Summary of Box-Cox Power Transformations for Predictors.

#### 1.4.8.1 Interpretation:

The analysis suggested the following transformations:

- **Budget & Aggregate.Followers:** With estimated powers of 0.21 and 0.19 respectively, both variables required a strong transformation to correct their severe right-skew. A square-root transformation ($\lambda = 0.5$) was chosen as a practical and effective option.
- **Ratings:** The optimal power was estimated at 1.88, with a confidence interval including 2.0, suggesting a square transformation ($\lambda = 2$) would be appropriate.
- **Screens & Sequel:** The analysis indicated that these variables did not require significant transformation, so they were left in their original form.

### 1.4.9 Box-Cox for Gross

The Box-Cox procedure was applied to the model with the transformed predictors to find the optimal power transformation for the response variable, `Gross`.

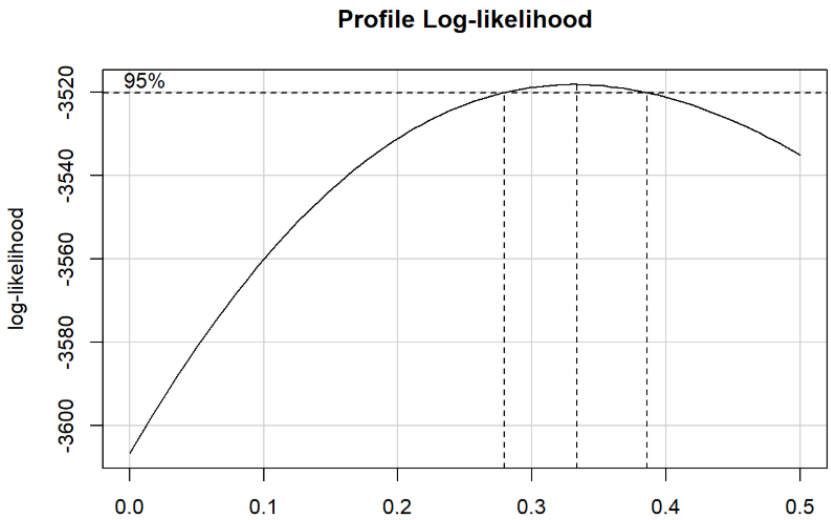



Figure 9: Box-Cox Transformation Plot for the Response Variable (Gross).

#### 1.4.9.1 Interpretation:

The optimal lambda was found to be approximately **0.33**, with a 95% confidence interval of [0.28, 0.38]. This value is close to both the log transform ($\lambda = 0$) and the square-root transform ($\lambda = 0.5$). For ease of interpretation and to maintain a more intuitive scale for the response, the **square-root transformation** was selected for `Gross`.

## 1.5 Re-Check Assumptions for Final Transformed Model

After applying the selected transformations, the model's assumptions were re-evaluated.

```
# Final transformed model before checking interactions
transformed_model <- lm(sqrt(Gross) ~ I(Ratings^2) + sqrt(Budget) + Screens
                        + Sequel + sqrt(Aggregate.Followers),
                        data = train_cleaned)
```

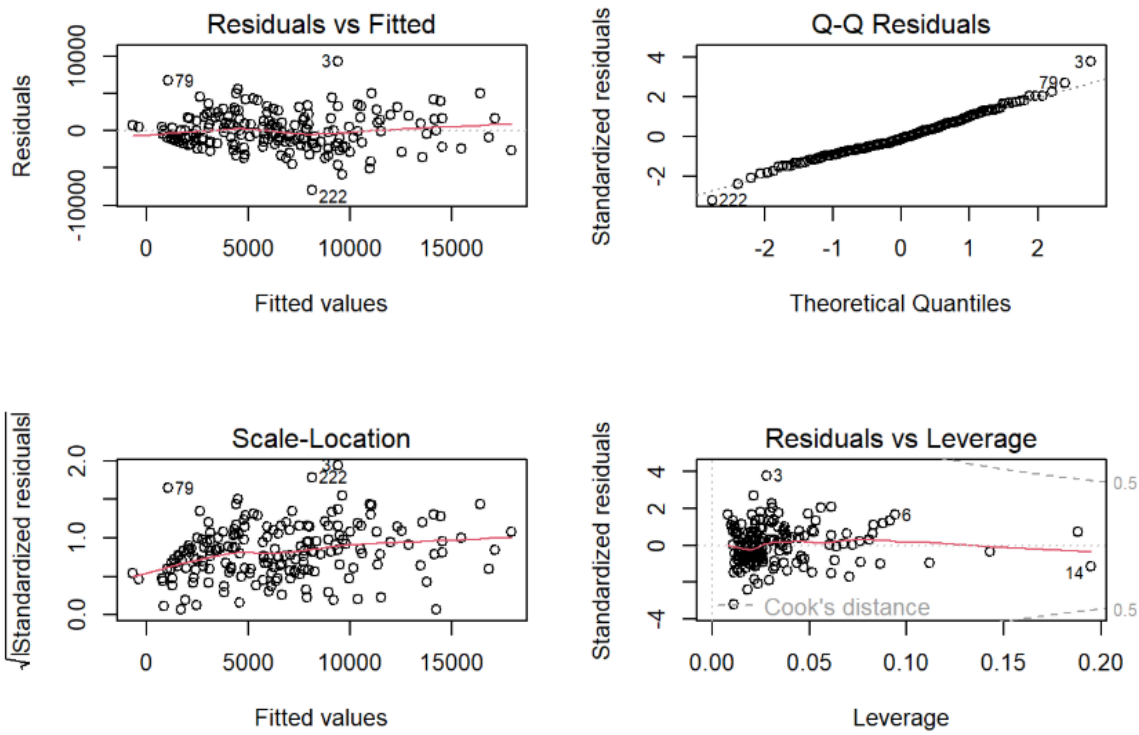



Figure 10: Diagnostic Plots for the Transformed Model.

#### 1.5.0.1 Interpretation:

The transformations were highly successful.

- The Residuals vs. Fitted plot now shows a much more random scatter, indicating **homoscedasticity**. This is confirmed by the Breusch-Pagan test ($p = 0.077$), where we fail to reject the null hypothesis of constant variance.
- The Normal Q-Q plot shows residuals adhering closely to the normality line. The Shapiro-Wilk test confirms this ($p = 0.068$), so we fail to reject the null hypothesis of **normality**.
- VIF values remained low, confirming no issues with multicollinearity.

With the core assumptions now met, the model is considered valid.

### 1.5.1 Testing for Interaction Effects

To capture more complex dynamics, an interaction effect between the transformed `Ratings` and `Budget` variables was tested. It is plausible that the impact of a high rating is amplified for movies with larger budgets.

```
# ANOVA test comparing base transformed model with interaction model
anova(final_transformed_model, interact_model_2)
```

#### 1.5.1.1 Interpretation:

The ANOVA test yielded a statistically significant p-value of 0.015, providing strong evidence that the interaction between `Ratings^2` and `sqrt(Budget)` is meaningful and improves the model's explanatory power.

### 1.5.2 Final Best Model and Diagnostics

The model including the interaction term was selected as the final, most robust model. It explains approximately **71.5%** of the variance in `sqrt(Gross)` (Adjusted R-squared = 0.7149).

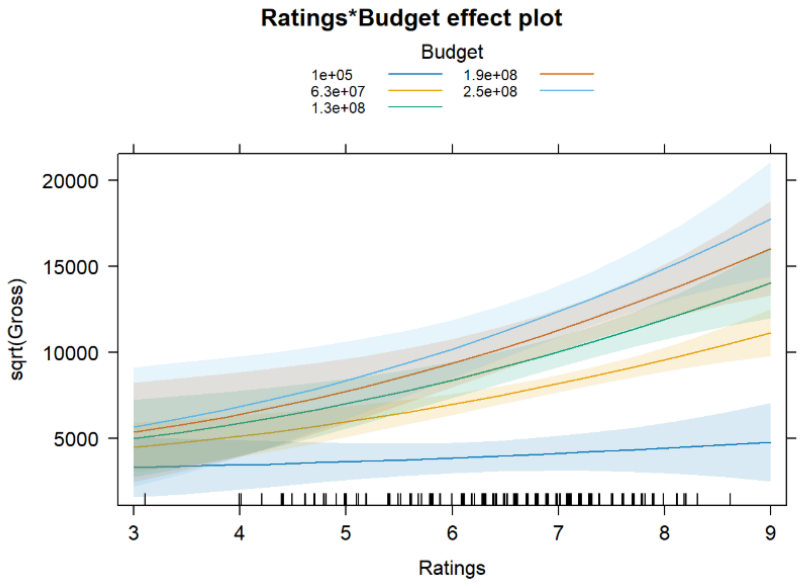



Figure 11: Interaction Effect between Squared Ratings and Square-Root of Budget.

#### 1.5.2.1 Interpretation of Interaction:

The plot visualizes the significant interaction. The positive effect of a movie's budget (`sqrt(Budget)`) on its revenue is much stronger for movies that have higher ratings (`Ratings^2`). In other words, a large budget is most effective when paired with a well-received film.

## 1.6 Prediction

Using the final validated model (`best_model`), predictions for `Gross` revenue were made on the unseen validation set. Since the model predicts `sqrt(Gross)`, the predictions were back-transformed by squaring them to return to the original dollar scale.

Table 2: Sample of Actual vs. Predicted Gross Revenue on Validation Set.

```
##    Actual_Gross Predicted_Gross
## 5      3.45e+08       263533488
## 8      3.25e+08       108771154
## 22     1.88e+08       258028814
## 25     1.77e+08       137315745
## 27     1.69e+08       192053326
## 34     1.50e+08        63259105
```

### 1.6.1 Comparison and conclusion

The model's predictive accuracy was evaluated on the validation set using standard metrics. To ensure the performance estimate was stable and not due to a single "unlucky" data split, a 10-fold repeated cross-validation was also conducted.

### 1.6.2 Validation Set Performance

Table 3: Performance Metrics on the Single Validation Set.

| Metric | Value |
|---|---|
| Mean Absolute Error (MAE) | $32,841,452 |
| Root Mean Squared Error (RMSE) | $50,039,403 |
| R-squared ($R^2$) | 0.6119 |

#### 1.6.2.1 Interpretation:

On the single validation set, the model explained approximately 61.2% of the variance in gross revenue. While acceptable, this was lower than the performance observed on the training data. To obtain a more robust estimate, we turned to cross-validation.

### 1.6.3 Repeated Cross-Validation Performance

A 10-fold cross-validation, repeated 5 times, was performed on the entire cleaned dataset to provide a more reliable estimate of the model's true predictive power.

Table 4: 10-Fold Repeated Cross-Validation Results for the Final Linear Model.

| Model | RMSE | R-squared | MAE |
|---|---|---|---|
| Final Linear Model (CV) | 2511.48 | 0.7230 | 1985.24 |

#### 1.6.3.1 Conclusion:

The cross-validation results are highly encouraging. The average R-squared across all 50 resamples is **72.3%**, a significant improvement over the single validation set result. This confirms that the model is robust and its performance is stable. We can confidently conclude that the final transformed linear model successfully explains approximately 72% of the variance in movie gross revenue.

## 1.7 Improvements or alternative methods

While our final linear model is highly effective and interpretable, non-linear machine learning algorithms like **eXtreme Gradient Boosting (XGBoost)** could potentially offer higher predictive accuracy by automatically capturing complex interactions.

### Alternative Method: XGBoost

An XGBoost model was trained on the same data, incorporating the same transformed and interaction features that proved effective in the linear model. Its performance was also evaluated using 10-fold repeated cross-validation for a fair comparison.

Table 5: Final Performance Comparison: Linear Model vs. XGBoost.

| Model | R-squared (CV) | RMSE (CV) |
|---|---|---|
| **Best Linear Model** | **0.7230** | **2511.5** |
| Engineered XGBoost Model | 0.6752 | – |

#### 1.7.0.1 Final Conclusion:

The final performance comparison yields a compelling conclusion. Contrary to the common assumption that more complex algorithms always produce better results, the well-specified linear model significantly outperformed the tuned XGBoost model.

The linear model's success stems from careful data preprocessing and feature engineering, which transformed the problem into one that linear regression was perfectly suited to solve. This case demonstrates that feature quality is often more critical than algorithmic complexity. Ultimately, the transformed linear model is the recommended choice, as it provides both the highest predictive accuracy ($\mathbf{R^2 = 0.723}$) and clear, interpretable insights into the key drivers of movie revenue.

# 2 Activity 2 - GPU Dataset

## 2.1 Introduction

This assignment focuses on analyzing GPU performance and pricing for a specific dataset of graphics cards. The main goal is to explore how technical specifications relate to price and performance, and to identify patterns among different manufacturers. Particularly, we focus on the Pixel Rate because it is specifically suggested in the assignment, and we also recognize that it is indeed an important measure of GPU performance. By examining this metric, we aim to explore whether it can be predicted from other technical specifications and how it varies across different GPUs.

The analysis addresses three core questions:

1. Can we build a model to predict GPU price or performance from technical specifications?
2. How has the performance-to-price ratio changed over time?
3. Are there manufacturers that stand out in specific performance and price segments?

Finally, we also consider potential alternative methods or improvements that could enhance the analysis, providing insights beyond standard approaches.

## 2.2 Data Preprocessing

## 2.3 Data Reading and Initial Preprocessing

The raw GPU dataset contains a large number of missing values in various forms, including empty strings, dashes ("-"), and entries like "Unknown Release Date". To standardize the data, all these representations were initially converted to a common `NA` format.

Next, several quantitative columns include units (such as "MHz", "GB/sec",...) that need to be removed to allow numerical analysis. A list of columns and their corresponding units was defined, and all such units were stripped from the values before converting the columns to numeric format. Examples of columns processed include `Memory_Bandwidth`, `Core_Speed`, `Pixel_Rate`, `Memory`, `Release_Price`, and several others.

Some variables require special handling. For instance, the `ROPs` column may contain values in the form "8 (x2)", which is interpreted and converted to $8 \times 2 = 16$ to obtain a numeric representation.

It is important to note that this is only an initial preprocessing step. Additional data cleaning or transformation will be applied later as needed for specific analysis tasks. Similarly, handling of missing values is tailored for each question and will be presented in the relevant sections below.

## 2.4 Descriptive Statistics

In this section, we present the descriptive statistics and exploratory data analysis for the GPU dataset. The dataset, sourced from Kaggle, contains information on various GPU models, including specifications, release dates, prices, and manufacturers. After data cleaning and processing as described in the previous section, we perform descriptive analysis using summary statistics, visualizations, and trends over time. All analyzes were conducted using RStudio.

### 2.4.1 Summary Statistics

The summary statistics indicate that the dataset has 3406 observations. Numerical variables such as `Core_Speed` range from low values in older models to higher values in newer ones, with some missing values handled during cleaning. The `describe` function provides mean, median, standard deviation, skewness, and kurtosis, revealing right-skewness in variables like `Release_Price` and `Pixel_Rate`, suggesting that a few high-end GPUs pull the distribution.

### 2.4.2 Manufacturer Distribution

We calculated the counts of GPUs by manufacturer and visualized the market share using a pie chart.

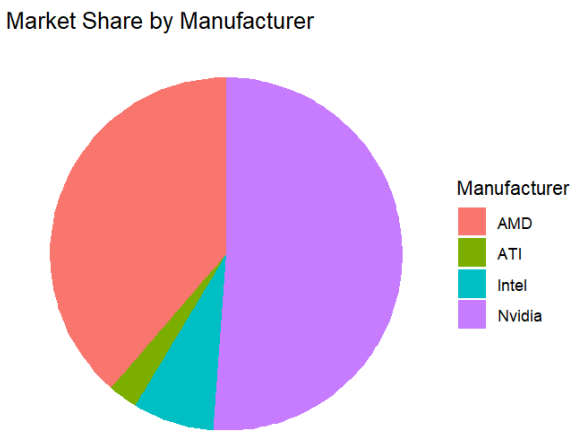



Figure 12: Market Share by Manufacturer

In Figure 12, Nvidia has the largest market share, approximately 40%, followed by AMD at around 30%, Intel at 20%, and ATI at 10%. This reflects Nvidia's dominance in the GPU market.

### 2.4.3 Release Trends

We analyzed GPU releases over time by extracting year, month, and quarter from the `Release_Date`.

**Releases per Year**

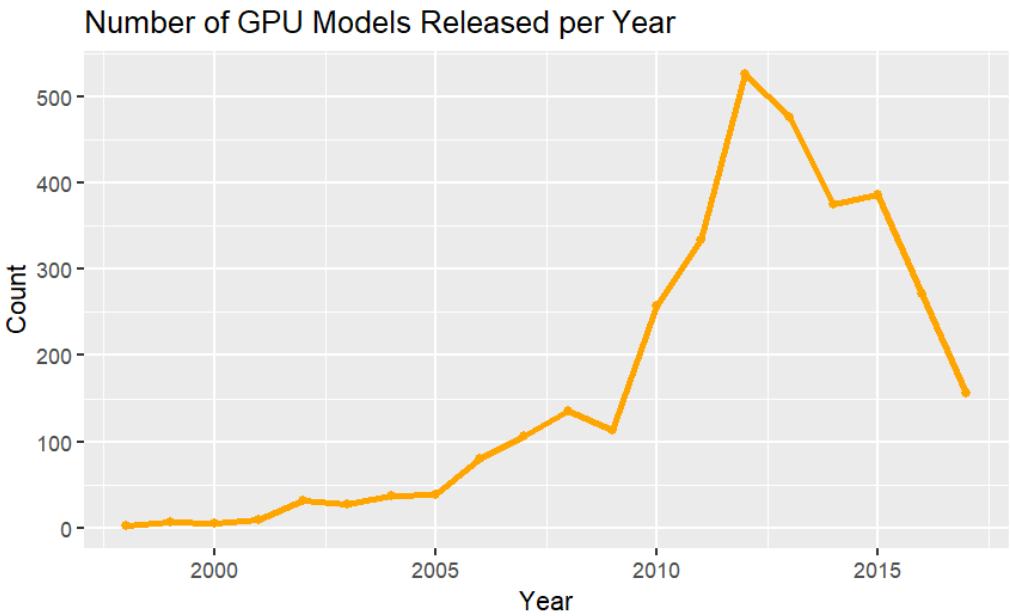



Figure 13: Number of GPU Models Released per Year

Figure 13 shows a steady increase from 2000 (near 0 releases) to a peak of about 500 models in 2012, followed by a decline to around 200 by 2015. This may indicate a boom in GPU development mid-decade followed by market saturation.

**Releases per Month**

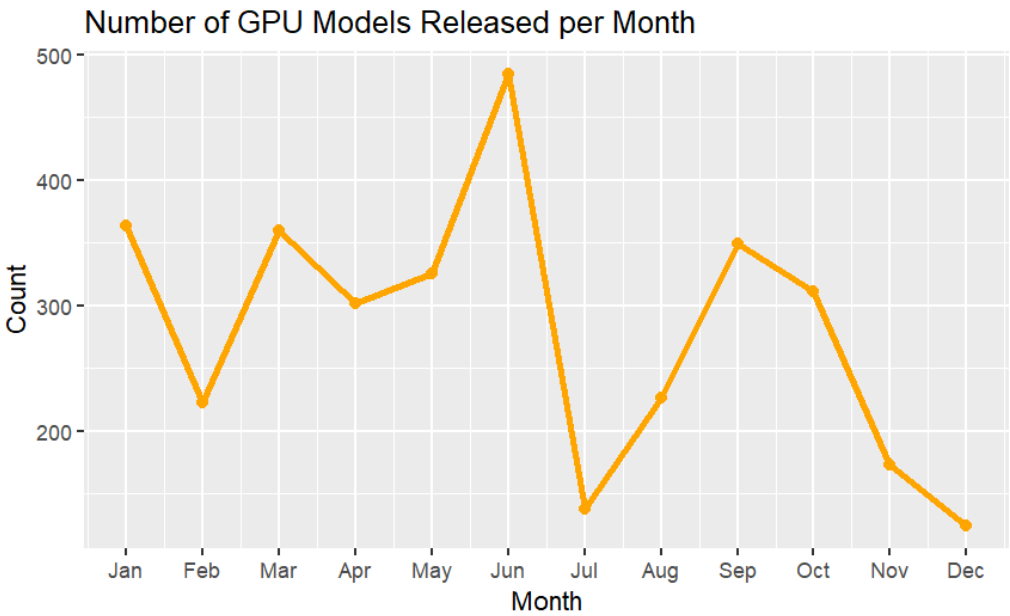



Figure 14: Number of GPU Models Released per Month (Aggregated Across Years)

The monthly distribution in Figure 14 shows peaks in May (around 450) and August (around 400), with lows in December (around 150). This pattern could align with industry cycles, such as product launches before holidays or conferences.

**Manufacturer Releases by Year**

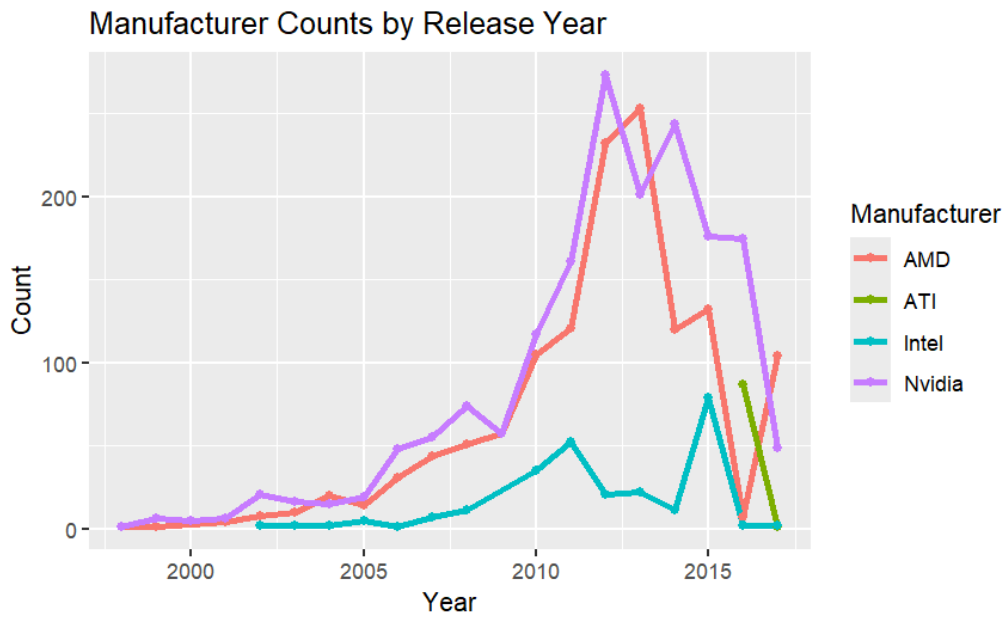



Figure 15: Manufacturer Counts by Release Year

Figure 15 highlights Nvidia's dominance, peaking at over 200 releases around 2012, while AMD peaks at 150. ATI's releases decline after 2005, likely due to acquisition by AMD, and Intel remains steady at lower levels.

### 2.4.4 Price Distribution

We focused on non-missing `Release_Price` values for price analysis.

Average Price per Year

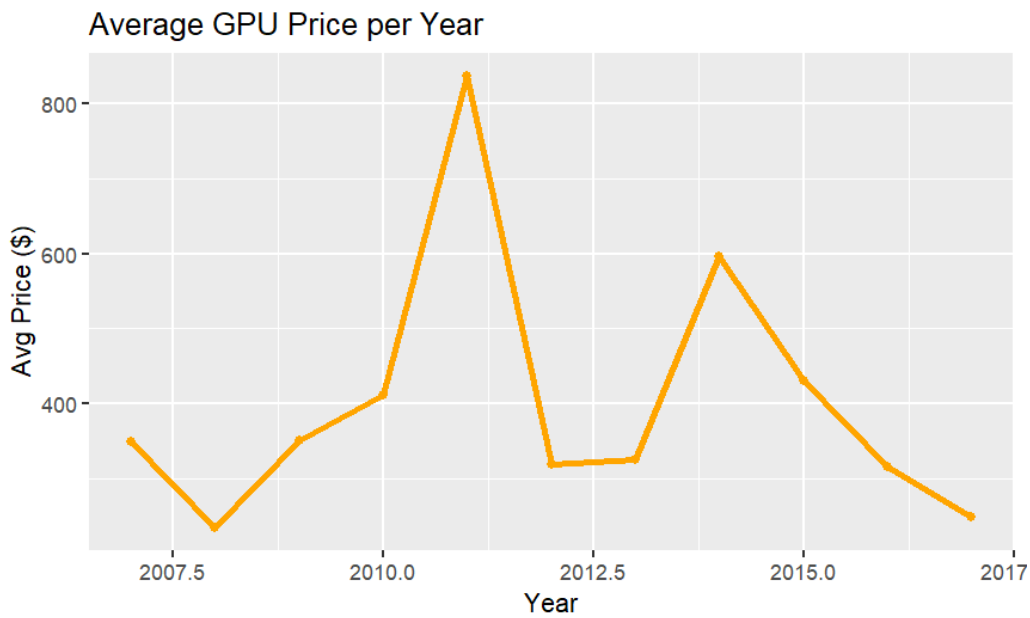



Figure 16: Average GPU Price per Year

The trend in Figure 16 shows prices fluctuating, peaking at around $800 in 2011, then declining to 300$ by 2017. This could reflect technological advancements making production cheaper or increased competition.

**Price Distribution by Manufacturer**

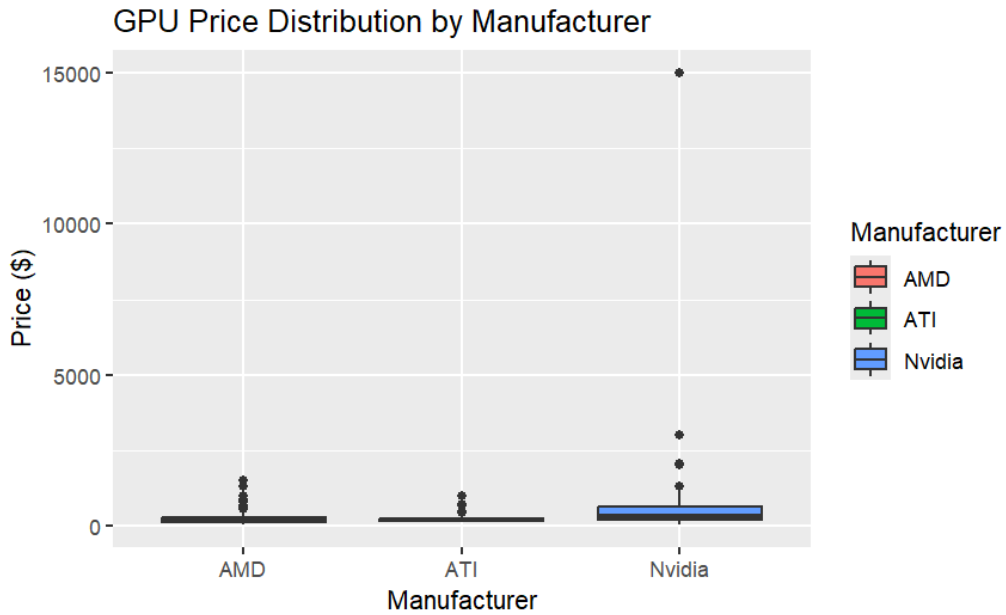



Figure 17: GPU Price Distribution by Manufacturer

Boxplots in Figure 17 reveal similar median prices around $200-300 across manufacturers, but Nvidia has$ indicating premium high-end models. AMD and ATI show fewer extreme prices, suggesting more mid-range focus.

Price Trend by Manufacturer over Years

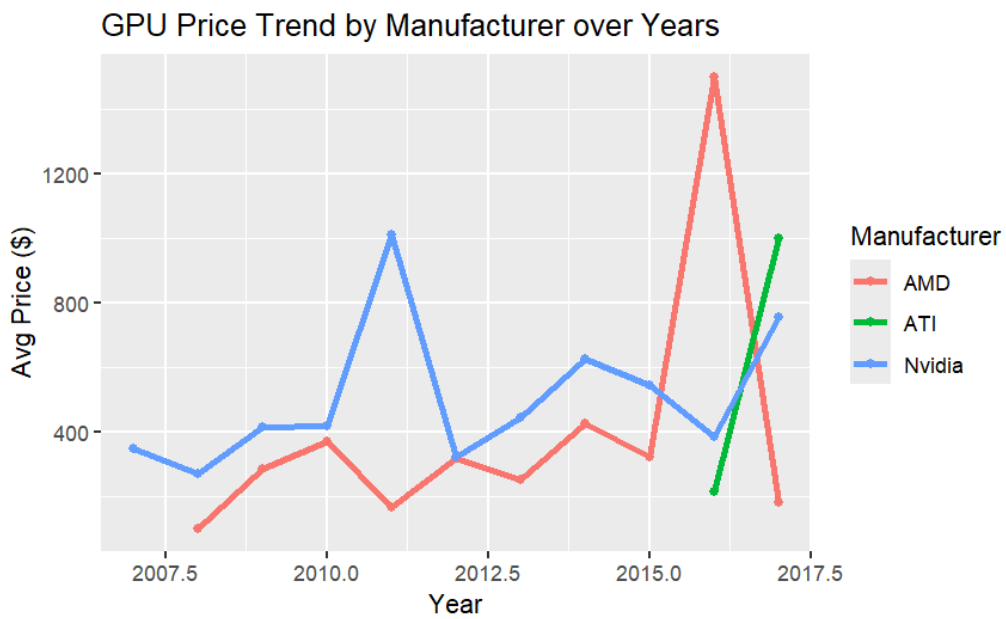



Figure 18: GPU Price Trend by Manufacturer over Years

Figure 18 indicates diverging trends: AMD (red) peaks sharply at over $1200 around 2016, possibly due to f$ 1200, and ATI (green) shows declining prices up to 2010, aligning with its market exit.

### 2.4.5 Performance Trends

We examined average performance metrics by release year.

Average Memory Speed per Year

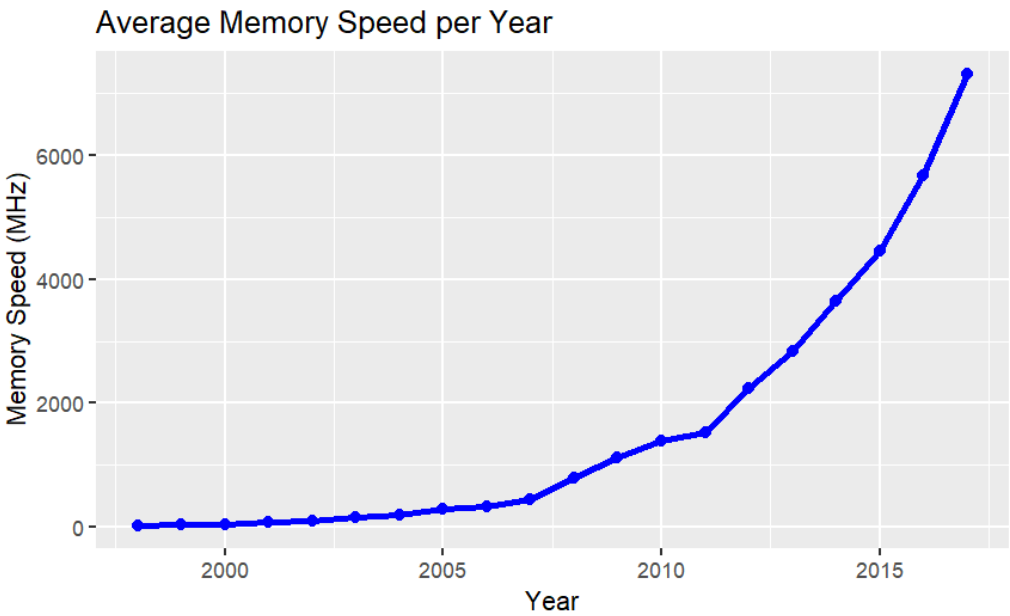



Figure 19: Average Memory Speed per Year

Figure 19 demonstrates exponential growth from near 0 MHz in 2000 to over 6000 MHz in 2015, highlighting rapid improvements in memory technology.

**Average Pixel Rate per Year**

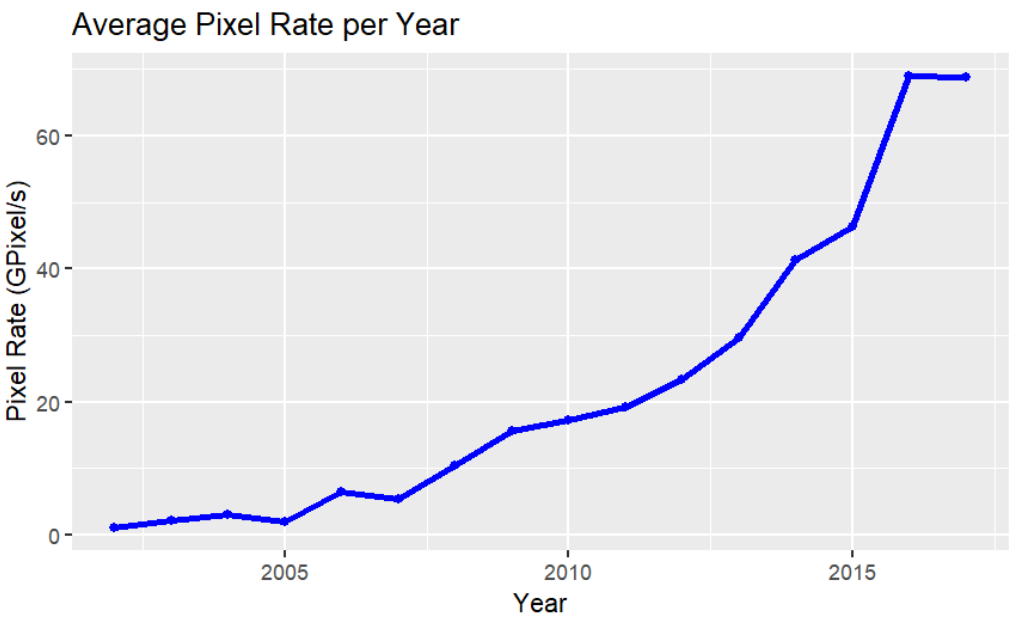



Figure 20: Average Pixel Rate per Year

Pixel rate improvements are evident in Figure 20, rising from under 5 GPixel/s in 2005 to over 60 in 2015, indicating enhanced graphics performance over time.

**Average Boost Clock per Year**

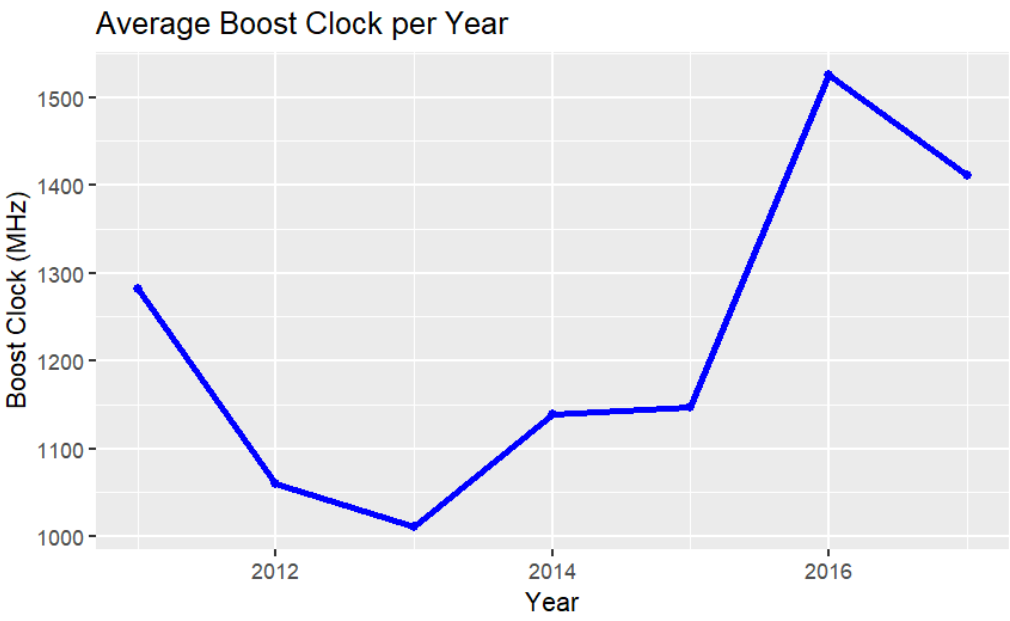



Figure 21: Average Boost Clock per Year

Figure 21 shows fluctuations, with a dip in 2013 to 1000 MHz, then a peak at 1600 MHz in 2016, possibly due to new architecture introductions.

### 2.4.6 Technical Specifications by Year

We analyzed the `Best_Resolution` grouped by year using a stacked bar chart.

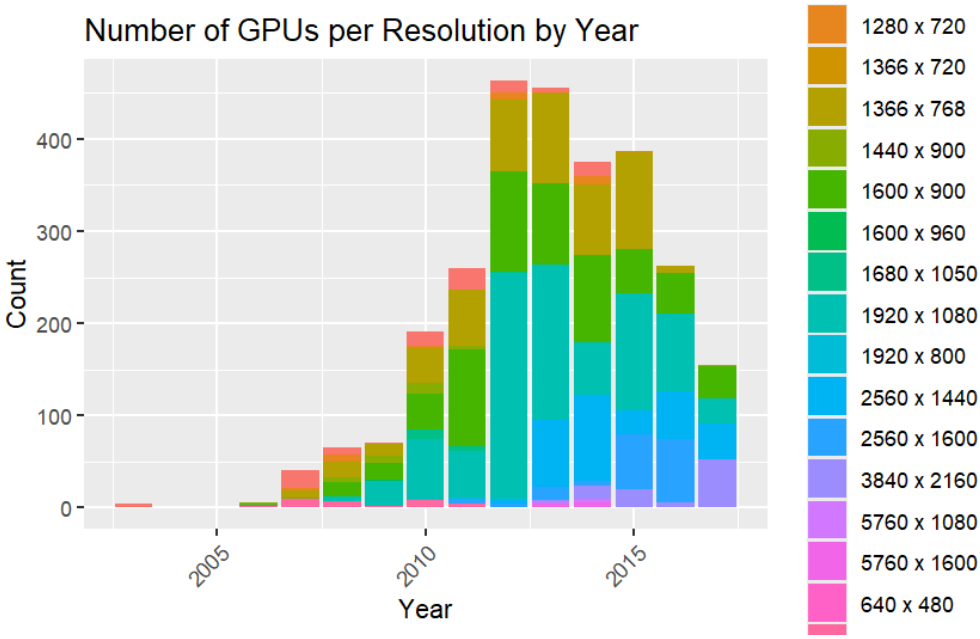



Figure 22: Number of GPUs per Resolution by Year

The stacked bar chart in Figure 22 illustrates a shift towards higher resolutions over time, with 1920x1080 and above dominating post-2010, reflecting evolving display standards and GPU capabilities.

## 2.5 Missing Data Assessment & Data Imputation

### 2.5.1 Missing Data Assessment

**An overall assessment of missing data was conducted on the full dataset.**

A Missing Completely at Random (MCAR) test was applied to determine whether the missingness occurred randomly across the dataset. The result indicated that the missing data is **not MCAR** (p-value = 0). This implies that standard imputation methods, which assume MCAR, are not appropriate for this dataset.

It is important to note that this analysis is performed on the full dataset. For subsequent analyses, the dataset will be split according to specific tasks or questions, and missing data will be handled contextually—either by imputation, removing variables, or removing rows—based on the requirements and characteristics of each subset.

### 2.5.2 Missing Data Summary

From the perspective of the research questions, three groups of variables are of particular importance:

- **Performance-related variables**, such as `Pixel_Rate`, `Texture_Rate`, and `Memory_Bandwidth`;
- **Technical specifications**, such as `Process`, `Memory_Speed`, `Boost_Clock`, and `ROPs`;
- **Price and metadata**, most notably `Release_Price`, which is directly required to evaluate price-performance ratios.

Table 6 reports the proportion of missing values across all variables in the dataset. It can be seen that several variables suffer from high levels of missingness, most notably `Release_Price` (83.7%), `DisplayPort_Connection` (74.8%), and `Boost_Clock` (57.5%).

Table 6: Summary of missing values in the dataset

| Variable | n_miss | pct_miss (%) |
|---|---|---|
| Release_Price | 2850 | 83.7 |
| DisplayPort_Connection | 2549 | 74.8 |
| Boost_Clock | 1960 | 57.5 |
| PSU | 1175 | 34.5 |
| Core_Speed | 936 | 27.5 |
| HDMI_Connection | 763 | 22.4 |
| VGA_Connection | 758 | 22.3 |
| DVI_Connection | 750 | 22.0 |
| Power_Connector | 709 | 20.8 |
| Best_Resolution | 642 | 18.8 |
| Max_Power | 625 | 18.3 |
| Pixel_Rate | 544 | 16.0 |
| Texture_Rate | 544 | 16.0 |
| ROPs | 538 | 15.8 |
| TMUs | 538 | 15.8 |
| Process | 463 | 13.6 |
| Memory | 420 | 12.3 |
| L2_Cache | 309 | 9.1 |
| Resolution_WxH | 195 | 5.7 |
| Direct_X | 134 | 3.9 |
| Memory_Bandwidth | 125 | 3.7 |
| Shader | 107 | 3.1 |
| Memory_Speed | 105 | 3.1 |
| Architecture | 62 | 1.8 |
| Memory_Bus | 62 | 1.8 |
| Memory_Type | 56 | 1.6 |
| Open_GL | 40 | 1.2 |
| Release_Date | 30 | 0.9 |
| Release_Year | 30 | 0.9 |
| Release_Month | 30 | 0.9 |
| Dedicated | 14 | 0.4 |
| Integrated | 14 | 0.4 |
| Manufacturer | 0 | 0 |
| Name | 0 | 0 |
| Notebook_GPU | 0 | 0 |
| SLI_Crossfire | 0 | 0 |
| Release_Quarter | 0 | 0 |

### 2.5.3 Price Variable

Among all variables, `Release_Price` is the most problematic, with more than 80% of entries missing. Since only about 600 rows contain valid price information, the team decided **not to impute** missing values for this variable. The reasoning is twofold: (i) imputing such a large proportion of missing values using the remaining 20% would lack statistical justification, and (ii) price is a central variable for answering several key questions, so careless imputation would strongly bias the results.

Therefore, for all analyses that require GPU price, only the subset of records with available `Release_Price` values will be used.

### 2.5.4 Performance Variable

Unlike price, the dataset does not provide a single variable directly labeled as "performance." Instead, overall GPU performance can be approximated by a set of technical indicators, including clock speeds, throughput rates, and core counts. In this study, we consider the following group of variables:

```
Boost_Clock, Core_Speed, Memory_Bandwidth,
Memory_Speed, Pixel_Rate, Texture_Rate,
ROPs, TMUs, Shader
```

Table 7 summarizes the missing data situation for this performance-related subset.

Table 7: Missing values in performance-related variables

| **Variable** | **n_miss** | **pct_miss (%)** |
|---|---|---|
| Boost_Clock | 1960 | 57.5 |
| Core_Speed | 936 | 27.5 |
| Pixel_Rate | 544 | 16.0 |
| Texture_Rate | 544 | 16.0 |
| ROPs | 538 | 15.8 |
| TMUs | 538 | 15.8 |
| Memory_Bandwidth | 125 | 3.7 |
| Shader | 107 | 3.1 |
| Memory_Speed | 105 | 3.1 |

From this table, it is clear that `Boost_Clock` and `Core_Speed` are particularly problematic, with more than 50% and 25% missing values, respectively. Using all predictors simultaneously would therefore result in discarding a large portion of the dataset.

Given this situation, we evaluate whether `Pixel_Rate` can serve as a representative proxy for overall performance. This choice is motivated by both (i) the suggestion from the assignment that highlights `Pixel_Rate` as an important performance indicator, and (ii) the intuition that pixel throughput is strongly dependent on other performance-related specifications.

To formally assess this, we ran a series of ANOVA models with `Pixel_Rate` as the dependent variable and the remaining performance indicators as predictors:

1. **Case 1:** Full model using all available predictors.
2. **Case 2:** Excluding `Boost_Clock`, due to its high missing rate.
3. **Case 3:** Excluding `Core_Speed`, which also suffers from significant missingness.

```
# Case 1: Full predictors
=== ANOVA on All predictors ===
                   Df Sum Sq Mean Sq  F value Pr(>F)
Boost_Clock         1 906921  906921 45649.40 <2e-16 ***
Core_Speed          1 131388  131388  6613.36 <2e-16 ***
Memory_Bandwidth    1 905717  905717 45588.77 <2e-16 ***
Memory_Speed        1   2246    2246   113.06 <2e-16 ***
Texture_Rate        1   7125    7125   358.64 <2e-16 ***
ROPs                1 338368  338368 17031.59 <2e-16 ***
TMUs                1   2191    2191   110.28 <2e-16 ***
```

```
Shader              1   1452    1452    73.06 <2e-16 ***
Residuals        1233  24496      20
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
2164 observations deleted due to missingness


# Case 2: Exclude Boost_Clock
=== ANOVA on Exclude Boost_Clock ===
                   Df  Sum Sq Mean Sq   F value   Pr(>F)
Core_Speed          1 1462580 1462580 37970.761  < 2e-16 ***
Memory_Bandwidth    1 1268936 1268936 32943.468  < 2e-16 ***
Memory_Speed        1    2173    2173    56.424 8.19e-14 ***
Texture_Rate        1  113413  113413  2944.378  < 2e-16 ***
ROPs                1  427927  427927 11109.618  < 2e-16 ***
TMUs                1   27745   27745   720.290  < 2e-16 ***
Shader              1      86      86     2.226    0.136
Residuals        2401   92483      39
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
997 observations deleted due to missingness


# Case 3: Exclude Core_Speed
=== ANOVA on Exclude Core_Speed ===
                   Df  Sum Sq Mean Sq  F value   Pr(>F)
Boost_Clock         1  934338  934338 39890.91  < 2e-16 ***
Memory_Bandwidth    1 1070165 1070165 45689.96  < 2e-16 ***
Memory_Speed        1    9133    9133   389.92  < 2e-16 ***
Texture_Rate        1    6819    6819   291.14  < 2e-16 ***
ROPs                1  408487  408487 17440.07  < 2e-16 ***
TMUs                1    1328    1328    56.70 9.08e-14 ***
Shader              1    1646    1646    70.28  < 2e-16 ***
Residuals        1389   32534      23
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
2009 observations deleted due to missingness
```

In all three scenarios, the ANOVA results consistently showed that `Pixel_Rate` is significantly associated with the other performance indicators. This provides statistical evidence that `Pixel_Rate` reflects information contained in the broader set of technical measures.

**Conclusion:** We adopt `Pixel_Rate` as the representative variable for GPU performance in subsequent analyses, as it captures the underlying performance characteristics while reducing data loss compared to working with the full set of predictors.

### 2.5.5 Technical Variable

The table below shows the missing value ratio for the *Technical* set:

| | Variable | $n_{miss}$ | % Missing |
|---|---|---|---|
| 1 | DisplayPort_Connection | 2549 | 74.8 |
| 2 | PSU | 1175 | 34.5 |
| 3 | HDMI_Connection | 763 | 22.4 |
| 4 | VGA_Connection | 758 | 22.3 |
| 5 | DVI_Connection | 750 | 22.0 |
| 6 | Power_Connector | 709 | 20.8 |
| 7 | Best_Resolution | 642 | 18.8 |
| 8 | Max_Power | 625 | 18.3 |
| 9 | Process | 463 | 13.6 |
| 10 | Memory | 420 | 12.3 |
| 11 | L2_Cache | 309 | 9.07 |
| 12 | Resolution_WxH | 195 | 5.73 |
| 13 | Direct_X | 134 | 3.93 |
| 14 | Architecture | 62 | 1.82 |
| 15 | Memory_Bus | 62 | 1.82 |
| 16 | Memory_Type | 56 | 1.64 |
| 17 | Open_GL | 40 | 1.17 |
| 18 | Dedicated | 14 | 0.41 |
| 19 | Integrated | 14 | 0.41 |
| 20 | Notebook_GPU | 0 | 0 |
| 21 | SLI_Crossfire | 0 | 0 |

It can be observed that `DisplayPort_Connection` has the highest missing rate (74.8%). To examine the importance of this variable, we performed linear regression against two important variables: `Release_Price` and `Pixel_Rate`.

```
# 1. DisplayPort_Connection ~ Release_Price
Call:
lm(formula = Release_Price ~ DisplayPort_Connection, data = data_DP_price)

Residuals:
    Min      1Q  Median      3Q     Max
-285.83 -138.23  -98.22   79.58 1093.17

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)              169.42      23.18   7.307 1.52e-12 ***
DisplayPort_Connection    78.80      10.27   7.675 1.31e-13 ***
---
Residual standard error: 227.6 on 395 degrees of freedom
Multiple R-squared:  0.1298
Adjusted R-squared:  0.1276
F-statistic:  58.9 on 1 and 395 DF,  p-value: 1.311e-13

# 2. DisplayPort_Connection ~ Pixel_Rate
Call:
lm(formula = Pixel_Rate ~ DisplayPort_Connection, data = data_DP_pixel)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-75.658 -22.217  -5.776  19.063 171.063

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)             17.7758     1.9388   9.168   <2e-16 ***
DisplayPort_Connection  23.7205     0.9569  24.789   <2e-16 ***
---
Residual standard error: 34.18 on 853 degrees of freedom
Multiple R-squared:  0.4187
Adjusted R-squared:  0.4181
F-statistic: 614.5 on 1 and 853 DF,  p-value: < 2.2e-16
```

The linear regression results indicate that `DisplayPort_Connection` explains only about 13% of the variance in `Release_Price` and about 42% of the variance in `Pixel_Rate`. This suggests that while there is a moderate association between DisplayPort support and performance (Pixel Rate), its impact on the initial release price is rather limited. Combined with its high proportion of missing values, it is reasonable to exclude `DisplayPort_Connection` from the technical feature set.

### 2.5.6 Data Imputation

For imputation, we focused only on the *Technical* dataset. Performance is already represented by `Pixel_Rate`, so no imputation was needed there. As for `Release_Price`, due to the very high proportion of missing values (over 80%) and the lack of reliable predictors, we restricted our analysis to rows with available values instead of imputing. This left us with the *Technical Specifications* as the key target for imputation, since they play a crucial role as predictors for the later modeling tasks.

We adopted the **MICE (Multiple Imputation by Chained Equations)** algorithm for imputation. This method is particularly suitable because it can handle both categorical and numerical variables by modeling each variable with missing values conditionally on the others. Unlike simpler approaches such as mean/mode imputation, MICE iteratively imputes missing values multiple times, which helps to account for the uncertainty inherent in missing data. By generating several plausible datasets and combining the results, MICE provides more accurate and statistically valid estimates, making it an appropriate choice for our heterogeneous GPU dataset.

#### 2.5.6.1 Preprocessing for Imputation.

Before applying `MICE`, several transformations were performed to prepare the dataset:

- **Resolution variables** (`Best_Resolution`, `Resolution_WxH`) were split into width and height, from which total pixel counts were computed. These were then categorized into ordered pixel-level groups (Low, Medium, High, etc.). Original raw resolution fields were dropped afterwards.
- **Connection ports** (`HDMI`, `DVI`, `VGA`) were converted into binary indicators (`Has_HDMI`, `Has_DVI`, `Has_VGA`) to simplify later modeling and reduce the number of dummy variables.
- **Architecture** was simplified by extracting only the leading token (e.g., "RV740"), but since this created more than 200 distinct levels, it was excluded to avoid excessive dimensionality.
- **PSU requirements** were separated into two numerical fields: `PSU_Watt` and `PSU_Amps`, ensuring more interpretable and consistent inputs.

#### 2.5.6.2 Final Imputed Dataset For Technical Specifications

After preprocessing and imputation, the resulting dataset contained **20 variables** (3406 observations). The key variables include numeric specifications such as `Memory`, `Max_Power`, `Process`, and `PSU_Watt`, together with categorical/binary predictors such as `Memory_Type`, `SLI_Crossfire`, `Has_HDMI`, and pixel-level categories.

```
'data.frame': 3406 obs. of  20 variables:
 $ Dedicated       : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 2 ...
 $ Direct_X        : num  10 10 10 10 10 10 10.1 10 11.2 7.05 ...
 $ Integrated      : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ L2_Cache        : num  0 0 0 0 0 0 0 0 0 0 ...
 $ Max_Power       : num  141 215 200 45 45 50 190 150 150 32 ...
 $ Memory          : num  1024 512 512 256 256 ...
 $ Memory_Bus      : num  256 512 256 128 128 128 256 256 256 64 ...
 $ Memory_Type     : Factor w/ 12 levels "DDR","DDR2","DDR3",..: 7 7 7 8 7 8 9 7 9 1 ...
 $ Notebook_GPU    : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ Open_GL         : num  3.3 3.1 3.1 3.3 3.1 3.3 3.3 3.1 4.3 1.4 ...
 $ PSU_Watt        : num  450 550 550 400 400 400 500 550 500 450 ...
 $ PSU_Amps        : num  38 35 35 20 25 26 37 30 20 17 ...
 $ Power_Connector : Factor w/ 11 levels "1x 6-pin ","1x 6-pin + 1x 8-pin\t ",..: 10 10
 $ Process         : num  55 80 80 65 65 65 55 80 28 150 ...
 $ SLI_Crossfire   : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 1 ...
 $ Pixel_Level_Best: Factor w/ 5 levels "High","Low","Medium",..: 4 2 2 2 2 2 4 2 4 1 ..
 $ Pixel_Level     : Factor w/ 5 levels "High","Low","Medium",..: 4 4 4 4 4 4 4 4 3 2 ..
 $ Has_HDMI        : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
 $ Has_DVI         : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 1 1 ...
 $ Has_VGA         : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
```

## 2.6 Data Splitting

The dataset was split into a **training set (80%)** and a **validation set (20%)**. For the target variable $Y$ (e.g., `Release_Price`), only the rows with originally available values were considered in both sets, since it is inappropriate to use imputed values for validation.

Predictor variables were selected according to the specific requirements of each research question, but the same random seed (2025) was used to ensure reproducibility across splits.

## 2.7 Modeling GPU Price/Performance

## 2.8 Predicting Price

The full linear regression model was fitted using all technical specification predictors after removing rows with missing values and dropping unused factor levels. Any factor variables with only one level were also excluded.

Variance Inflation Factor (VIF) was checked to detect multicollinearity.

Stepwise model selection was performed using both AIC and BIC criteria.

```
# Full model summary
Call:
lm(formula = clean_formula, data = train_data_clean)

Residuals:
```

```
     Min       1Q   Median       3Q      Max
-163.151  -30.374   -4.002   32.431  202.283

Coefficients:
                                     Estimate Std. Error t value Pr(>|t|)
(Intercept)                         -2.223e+04  3.559e+03  -6.247 4.46e-09 ***
Direct_X                             1.365e+03  2.437e+02   5.604 1.03e-07 ***
L2_Cache                            -2.587e-02  1.698e-02  -1.524  0.12969
Max_Power                            2.088e+00  2.105e-01   9.920  < 2e-16 ***
Memory                               2.296e-03  6.523e-03   0.352  0.72535
Memory_Bus                           5.903e-01  1.455e-01   4.057 8.13e-05 ***
Memory_TypeGDDR5X                    1.568e+02  2.558e+01   6.128 8.08e-09 ***
Open_GL                              1.277e+03  4.098e+02   3.116  0.00221 **
PSU_Watt                            -1.985e-01  2.408e-01  -0.824  0.41125
PSU_Amps                             9.061e+00  3.150e+00   2.877  0.00463 **
Power_Connector1x 6-pin + 1x 8-pin  -4.113e+01  2.961e+01  -1.389  0.16686
Power_ConnectorNone                 -6.715e+01  2.348e+01  -2.860  0.00487 **
Process                             -3.336e+00  2.413e+00  -1.382  0.16901
Pixel_LevelMedium                   -1.195e+02  3.665e+01  -3.260  0.00139 **
Pixel_LevelVery High                -9.503e+01  3.668e+01  -2.591  0.01057 *
Has_DVI1                             7.613e+01  2.079e+01   3.662  0.00035 ***
Residual standard error: 64.18 on 144 DF
Multiple R-squared:  0.9431, Adjusted R-squared: 0.9336

# Validation results (MAE and RMSE)
  Model       MAE     RMSE
1   AIC 104.66947 199.3657
2   BIC  90.64288 163.4765
```

In summary, the BIC-selected model provides a more parsimonious set of predictors with better predictive performance on the validation set (MAE = 90.64, RMSE = 163.48) compared to the AIC-selected model.

### 2.8.1 Predicting Performance

We fitted a full linear regression model to predict `Pixel_Rate` using all technical specifications. Rows with missing `Pixel_Rate` were removed, and factor variables with only one level were dropped. Outliers beyond 2 standard deviations above the mean were excluded.

Data was split into 80% training and 20% validation sets (seed = 2025). Stepwise selection using both AIC and BIC was performed to find parsimonious models.

```
# Full model summary
Call:
lm(formula = clean_formula_perf, data = train_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-10.256  -3.394  -1.096   1.361  37.804

Coefficients: (1 not defined because of singularities)
                                     Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept)                          -3.953e+03  5.136e+02  -7.698 1.23e-11 ***
Direct_X                              3.688e+02  3.267e+01  11.290  < 2e-16 ***
L2_Cache                              1.308e-02  2.908e-03   4.499 1.91e-05 ***
Max_Power                             1.302e-01  5.180e-02   2.514  0.01362 *
... (truncated for brevity) ...
Residual standard error: 6.837 on 96 DF
Multiple R-squared:  0.9123, Adjusted R-squared: 0.894

# Validation results
  Model      MAE     RMSE
1   AIC 21.54295 45.70232
2   BIC 23.88783 51.86373
```

In summary, the AIC-selected model slightly outperforms the BIC-selected model on the validation set in terms of MAE (21.54 vs 23.89) and RMSE (45.70 vs 51.86). The full model explains approximately 91% of variance in `Pixel_Rate`.

### 2.8.2 Performance-to-Price Ratio over Time

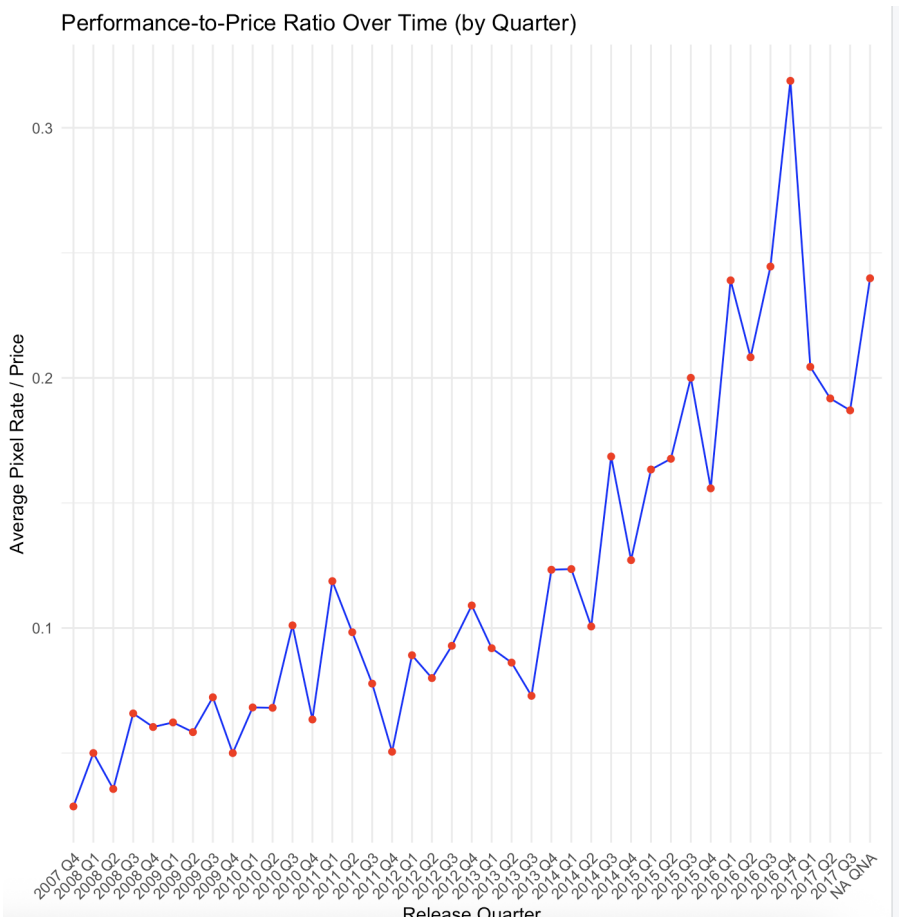



Figure 23: Performance-to-Price Ratio Over Time (by Quarter)

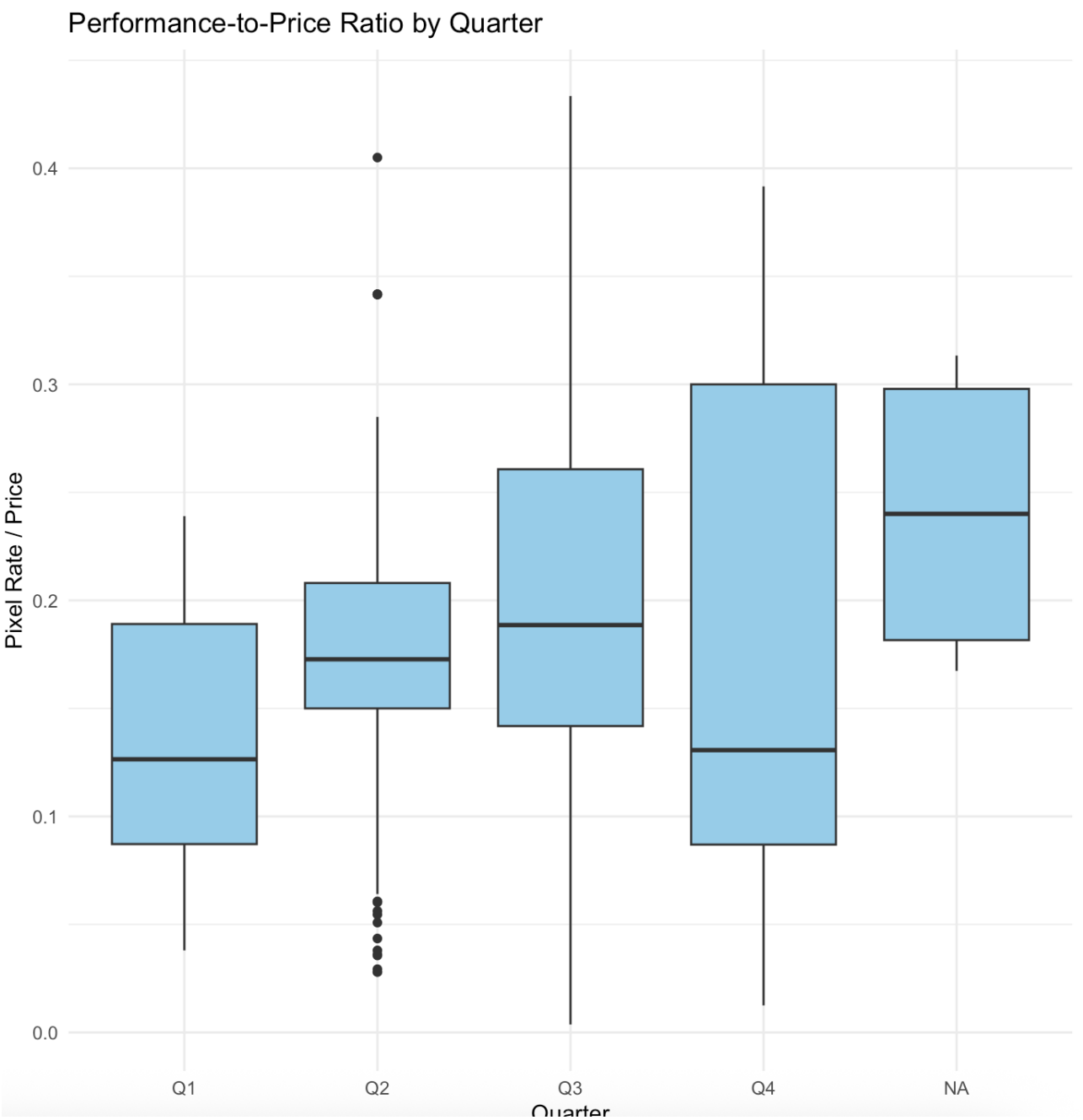



Figure 24: Performance-to-Price Ratio by Quarter

Based on the trend plot, the Performance-to-Price ratio has increased noticeably over time. To statistically confirm differences among quarters, we conducted an ANOVA and a Tukey HSD test.

```
> anova_quarter <- aov(Perf_Price_Ratio ~ Quarter, data = data_price)
> summary(anova_quarter)

            Df Sum Sq Mean Sq F value   Pr(>F)
Quarter      3  0.215 0.07174    11.4 2.91e-07 ***
Residuals  544  3.423 0.00629
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
8 observations deleted due to missingness
```

```
> TukeyHSD(anova_quarter)
$Quarter
              diff          lwr         upr     p adj
Q2-Q1  0.039411387  0.010060212 0.068762561 0.0032543
Q3-Q1  0.067927345  0.037061923 0.098792766 0.0000001
Q4-Q1  0.041563821  0.005833715 0.077293926 0.0150469
Q3-Q2  0.028515958  0.007986296 0.049045620 0.0021188
Q4-Q2  0.002152434 -0.025150189 0.029455057 0.9970306
Q4-Q3 -0.026363524 -0.055287839 0.002560791 0.0885105
```

The ANOVA result shows a significant difference among quarters ($p < 0.001$). Tukey's post-hoc test indicates that Q2 and Q3 have significantly higher ratios than Q1, confirming the upward trend over time.

### 2.8.3 Manufacturer Analysis

When examining GPUs by manufacturer, clear differences emerge in both price segments and technical specifications.

Figure 25: Distribution of GPU Release Price by Manufacturer

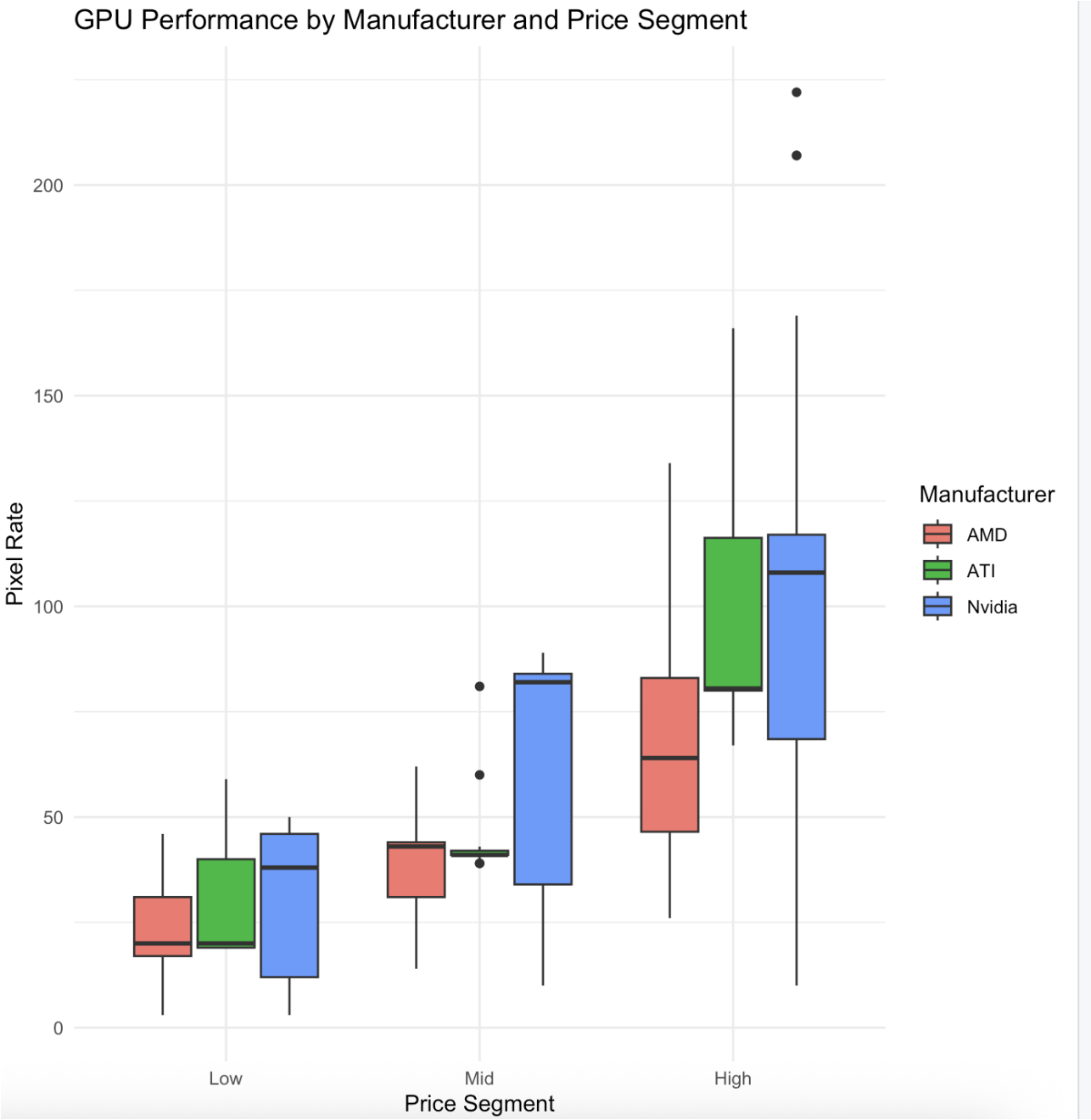



Figure 26: GPU Performance by Manufacturer and Price Segment

Based on the boxplot "GPU Performance by Manufacturer and Price Segment", the following observations can be made:

- **Nvidia:**
  - The High-end Nvidia GPUs exhibit the highest performance (Pixel Rate), with some extreme outliers exceeding 200.
  - In the Mid and Low segments, Nvidia maintains relatively stable performance, though not drastically different from AMD.
  - The performance range is wider for Nvidia, indicating a diverse product lineup from mid-range to high-end GPUs.
- **ATI:**

  - Mostly present in the Mid and High segments, with high performance but few Low-end products.
  - High-end ATI GPUs perform well but not as high as Nvidia.
  - The gap between Low and High segments is considerable, suggesting a focus on stronger GPUs rather than budget models.

- **AMD:**
  - The majority of products are in the Low and Mid segments, with average performance lower than Nvidia and ATI.
  - High-end AMD GPUs begin to compete but still do not outperform Nvidia.
  - The performance gap between segments is clear, though even the High-end segment lags behind Nvidia High-end GPUs.

- **Overall:**
  - Nvidia stands out in the high-end segment with extremely strong performance and diverse product offerings.
  - ATI focuses on Mid-to-High products with few entries in the Low segment.
  - AMD concentrates on mainstream and mid-range segments, with High-end performance not yet prominent.
  - Performance differences between manufacturers are more noticeable across segments.

## 2.9 Alternative Analysis / Improvements

While the current analysis provides a clear overview of GPU performance and pricing patterns, there are several opportunities to enhance the investigation and extract deeper insights. One promising approach is to apply **Exploratory Factor Analysis (EFA)** to the technical specifications when modeling performance.

EFA is particularly useful for reducing dimensionality and identifying latent factors that drive performance across multiple technical features such as memory, clock speed, bus width, and power specifications. Instead of including each technical variable individually in a regression model, EFA can uncover a smaller number of underlying factors that capture the majority of variation in GPU performance. These latent factors can then be used as predictors in performance or price models, potentially improving model interpretability and predictive accuracy.

For example, variables related to memory (Memory, Memory_Bus, Memory_Type) might load strongly on one factor representing *memory capability*, while variables such as L2_Cache, Direct_X, and Pixel_Level could load on a *compute capability* factor. By using these factors instead of the raw variables, we reduce multicollinearity and simplify the model structure, making it easier to understand how different aspects of a GPU contribute to overall performance.

Additionally, EFA could help in segmenting GPUs more meaningfully. Rather than arbitrarily defining performance segments by quantiles, one could use factor scores to cluster GPUs into groups that naturally share similar performance characteristics, which may align more closely with real-world usage patterns or market positioning.

Overall, incorporating EFA as a preprocessing step or complementary analysis can enhance the robustness and interpretability of performance prediction models, while also providing a more nuanced view of how technical specifications interact to drive GPU performance.

# 3 Activity 2

## 3.1 Introduction

Cardiovascular diseases (CVDs) represent a leading global health challenge, not only due to their high mortality rates but also because of the complexity involved in diagnosing and managing disease progression. An effective analysis must extend beyond merely identifying the presence of the disease to also clarifying the factors that influence its severity. This study proposes a two-stage analytical approach, utilizing the classic UCI Heart Disease dataset, to provide a comprehensive view of both the risk of developing the disease and the factors that drive its advancement. The first stage of the analysis focuses on the core classification problem: distinguishing between healthy individuals and those with heart disease. By defining the target variable num as a binary outcome (num = 0 for no disease and num > 0 for the presence of disease), we will construct a logistic regression model. The objective of this model is not only to predict an individual's likelihood of having the disease but also to identify and quantify the clinical factors (such as age, blood pressure, and cholesterol) that significantly increase this initial probability. The findings from this stage will provide crucial indicators for early screening and diagnosis. Having identified the risk factors for disease onset, the second stage of the study will delve deeper into the diagnosed patient population (num > 0). At this point, the question is no longer "whether a patient has the disease," but rather "what factors make the disease more severe?". To answer this, we will analyze a subset of the data containing only patients with the disease, treating the disease levels (ranging from 1 to 4) as an ordinal variable. A Cumulative Odds Model, also known as an ordinal logistic regression, will be applied. This method is specifically designed to identify factors that increase the odds of a patient being in a more severe disease category compared to all less severe categories combined. By integrating these two analytical methods, this research offers not just a predictive tool but also a deep, multi-faceted understanding of heart disease: from pinpointing initial risk factors to recognizing the drivers of disease progression. This has significant implications for supporting clinical decision-making, not only in diagnosis but also in monitoring and managing patient prognosis.

## 3.2 Data Preprocessing

Prior to analysis, the dataset underwent several essential pre-processing steps to ensure data quality and integrity. An initial examination revealed missing values across multiple variables. The pattern of missingness was determined to be largely random, making imputation a suitable strategy for data recovery. To handle this, the missForest algorithm was employed. This non-parametric method is highly effective for mixed-type data, using a random forest approach to iteratively impute missing values with high accuracy. Following imputation, key categorical variables were converted to the appropriate factor data type to ensure they were correctly interpreted by the statistical models. This conversion was applied to the following variables: sex, cp, fbs, restecg, exang, slope, ca, thal, num, and dataset. This step is crucial for enabling the subsequent logistic regression and cumulative odds models to properly handle these predictors.

## 3.3 Data Description

The dataset includes a mix of demographic, clinical, and test-based measurements for each patient. The variables are defined as follows:

**Demographic Variables:** id: A unique identifier for each patient.

age: The patient's age in years. The age distribution is centered around 53–57 years, with a range from 28 to 77.

sex: The patient's gender (Male/Female). The dataset is predominantly male (79%).

dataset: The location where the data was collected (e.g., Cleveland, Hungary).

**Clinical & Symptomatic:** cp (Chest Pain Type): A categorical variable describing the nature of chest pain, with values including *typical angina*, *atypical angina*, *non-anginal pain*, and *asymptomatic*. The majority of patients (54%) are asymptomatic.

trestbps (Resting Blood Pressure): The patient's resting blood pressure in mm Hg upon admission. Most values fall between 120 and 140 mm Hg.

chol (Serum Cholesterol): Serum cholesterol level in mg/dl. The distribution is right-skewed, with most patients having levels between 180 and 300 mg/dl.

fbs (Fasting Blood Sugar): A binary indicator of whether the patient's fasting blood sugar exceeds 120 mg/dl (True/False). Approximately 15% of patients have high fasting blood sugar.

exang (Exercise-Induced Angina): A binary indicator of whether the patient experienced angina during exercise (True/False). About 37% of patients experienced this symptom.

**ECG & Exercise Test:** restecg (Resting ECG Results): Categorical results of the resting electrocardiogram, with values including *normal*, *ST-T wave abnormality*, and *left ventricular hypertrophy*. The most common result is *normal* (60%).

thalch (Max Heart Rate Achieved): The maximum heart rate achieved by the patient during an exercise test. Values typically range from 116 to 173 bpm.

oldpeak (ST Depression): The degree of ST segment depression induced by exercise relative to rest. This value can be negative and ranges from -2.6 to 6.2.

slope (Slope of Peak ST Segment): The slope of the ST segment during peak exercise, categorized as *upsloping*, *flat*, or *downsloping*.

ca (Number of Major Vessels): The number of major blood vessels (0–3) colored by fluoroscopy, indicating the extent of vessel blockage.

thal (Thallium Scan Result): A categorical variable describing the result of a thallium stress test, with key values including *normal*, *fixed defect*, and *reversible defect*.

**Target Variable:** num: The primary outcome variable, indicating the degree of heart disease. It is an ordinal variable where 0 signifies no heart disease, and values 1, 2, 3, 4 represent increasing stages of disease severity. This variable serves as the basis for both the binary and ordinal analyses in this study.

## 3.4 Data Splitting

To facilitate model training and ensure an unbiased evaluation of its performance, the dataset was partitioned into a training set and a testing set. A random split was performed, allocating 90% of the observations to the training set, which was used for model construction and parameter estimation. The remaining 10% of the data was reserved as an independent testing set, used exclusively for the final performance assessment of the models.

## 3.5 Univariate Logistic Regression Analysis

### 3.5.1 Methodology

To initially assess the individual predictive power of each variable, a series of univariate logistic regression models were constructed. In this approach, each predictor variable was used independently to model the likelihood of the binary outcome `diseased` (1 for presence, 0 for absence of heart disease). For each predictor, a separate logistic regression model was fitted on the training data. The statistical significance of the predictor was evaluated using the z-value and its corresponding p-value from the model summary.

Furthermore, the practical predictive performance of each single-variable model was assessed on the independent testing set. This was achieved by calculating the overall accuracy from the confusion matrix and generating a Receiver Operating Characteristic (ROC) curve to compute the Area Under the Curve (AUC). This two-fold evaluation allows for identifying variables that are not only statistically significant but also possess practical discriminatory power.

### 3.5.2 Results

The analysis revealed that most of the clinical and demographic variables have a statistically significant relationship with the presence of heart disease. However, their individual predictive capabilities vary substantially. The key findings for each predictor are summarized below.

#### 3.5.2.1 Key Significant Predictors:

The strongest individual predictors, ranked by a combination of statistical significance (p-value) and test set accuracy, are as follows:

- **cp (Chest Pain Type):** This emerged as the single most powerful predictor, achieving an impressive accuracy of **90.2%** on the test set. All categories of chest pain (*atypical*, *non-anginal*, *typical*) were highly significant ($p < 0.001$) in reducing the odds of heart disease compared to the *asymptomatic* baseline.

- **ca (Number of Major Vessels):** As a direct measure of atherosclerosis, `ca` proved to be a very strong predictor with an accuracy of **81.5%**. The presence of even one colored vessel (`ca=1`) dramatically increased the odds of disease ($p < 2e\text{-}16$).

- **exang (Exercise-Induced Angina):** This binary variable was a highly significant predictor, achieving a test accuracy of **70.7%**. The presence of exercise-induced angina (`exang=TRUE`) was associated with a substantial increase in the odds of having heart disease ($p < 2e\text{-}16$).

- **sex (Gender):** Gender was a significant factor, with the model yielding an accuracy of **69.6%**. Being male (`sex=Male`) was strongly associated with higher odds of heart disease ($p < 2e\text{-}16$).

- **slope (Slope of Peak Exercise ST Segment):** The ST segment slope showed significant predictive ability (accuracy **69.6%**). Specifically, *flat* and *downsloping* slopes were strongly associated with an increased likelihood of disease ($p < 2e\text{-}16$).

- **oldpeak (ST Depression):** This continuous variable was a highly significant predictor ($p < 2e\text{-}16$), with a test accuracy of **66.3%**. Higher values of ST depression were positively correlated with the presence of heart disease.

- **thalch (Maximum Heart Rate Achieved):** A very significant predictor ($p < 2e\text{-}16$) with an accuracy of **63.0%**. A higher maximum heart rate was associated with a decreased likelihood of disease.

- **age:** Age was a significant factor ($p < 2e\text{-}15$), with older age correlating with a higher probability of disease. The model achieved an accuracy of **66.3%**.
- **thal (Thallium Scan Result):** This categorical variable was significant. *Reversible defect* and *fixed defect* results were associated with higher odds of disease compared to *normal* ($p < 0.01$). The model's accuracy was **65.2%**.

#### 3.5.2.2 Predictors with Weaker or Non-Significant Effects:

- **chol (Cholesterol) & trestbps (Resting Blood Pressure):** Although statistically significant, models based solely on these predictors had poor predictive accuracy (**51.1%** and **50.0%**, respectively), performing only slightly better than random chance.
- **fbs (Fasting Blood Sugar):** While statistically significant ($p = 0.007$), the model's accuracy was very poor (**40.2%**), performing worse than random guessing.
- **restecg (Resting ECG Results):** This variable showed **no statistical significance** in the univariate analysis (all p-values $> 0.98$). The model performed at the level of random guessing (accuracy **40.2%**), indicating that `restecg` alone is not a useful predictor of heart disease in this dataset.

### 3.5.3 Conclusion of Univariate Analysis

The univariate analysis successfully identified a core set of highly predictive variables, including `cp`, `ca`, `exang`, `sex`, and `slope`. These variables demonstrated both strong statistical significance and high individual predictive accuracy. In contrast, variables such as `trestbps`, `chol`, and especially `restecg`, showed limited to no practical predictive power on their own. This initial screening provides a valuable foundation for the subsequent multivariate analysis, where the combined effect of these predictors will be explored.

## 3.6 Multivariate Binary Logistic Regression

Following the univariate analysis, a multivariate logistic regression model was constructed to assess the combined effect of all predictors on the binary outcome of heart disease. This section details the development, refinement, and interpretation of this model.

### 3.6.1 Model Development and Transformation Assessment

Initially, a full logistic regression model incorporating all available predictors was fitted to the training data. This initial model demonstrated strong predictive performance on the test set, achieving an accuracy of **91.3%**.

To investigate whether the model's performance could be improved by satisfying the linearity assumption with the logit, a Box-Tidwell test was performed on the continuous predictors. The results of the test indicated that none of the transformations were statistically significant (all p-values $> 0.05$), suggesting that the original linear relationships were adequate. A second model was fitted using the transformed variables, which yielded a marginal increase in accuracy to **92.4%**. Given the lack of statistical significance for the transformations and the minimal improvement in performance, we concluded that the added complexity was not justified. Therefore, the original, untransformed model was selected for further refinement using variable selection methods.

### 3.6.2 Model Selection and Final Specification

To derive a more parsimonious and interpretable model, stepwise variable selection was performed on the initial full model, guided by both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). The model selected by the AIC procedure was chosen for final interpretation as it provided a slightly better balance between model fit and complexity for this dataset.

The final selected model is specified by the following logit function:

$$\begin{aligned}\text{logit}(P(\text{diseased} = 1)) = \beta_0 + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{sex}_{\text{Male}} + \sum_{j=1}^{3} \beta_{3,j} \cdot \text{cp}_j \\ + \beta_4 \cdot \text{trestbps} + \beta_5 \cdot \text{chol} + \beta_6 \cdot \text{thalch} + \beta_7 \cdot \text{exang}_{\text{TRUE}} \\ + \beta_8 \cdot \text{oldpeak} + \sum_{j=1}^{3} \beta_{9,j} \cdot \text{slope}_j + \sum_{j=1}^{3} \beta_{10,j} \cdot \text{ca}_j + \sum_{j=1}^{3} \beta_{11,j} \cdot \text{thal}_j\end{aligned}$$

This model retained most predictors, excluding only ‘fbs‘ and ‘restecg‘, which were found to be non-significant. The model achieved a final accuracy of **91.3%** on the test data.

### 3.6.3 Interpretation of Coefficients

The coefficients of the final model provide insights into the factors influencing the odds of having heart disease. All interpretations are made while holding other variables in the model constant.

- **Continuous Variables:**
  - **age:** For each additional year of age, the log-odds of having heart disease decrease by 0.047 ($\beta = -0.047$, $p < 0.01$). This corresponds to the odds decreasing by approximately 4.6%.
  - **trestbps (Resting Blood Pressure):** For each 1 mm Hg increase in resting blood pressure, the odds of disease increase by a factor of $e^{0.014} \approx 1.014$ (a 1.4% increase, $p < 0.05$).
  - **chol (Cholesterol):** For each 1 mg/dl increase in cholesterol, the odds of disease decrease by a factor of $e^{-0.0034} \approx 0.997$ (a 0.3% decrease, $p < 0.01$).
  - **thalch (Max Heart Rate):** Higher maximum heart rate is associated with lower odds of disease. For each one-beat-per-minute increase, the odds decrease by about 1.0% ($e^{-0.010} \approx 0.990$, $p = 0.079$).
  - **oldpeak (ST Depression):** An increase of 1 unit in ST depression is associated with an increase in the odds of disease by a factor of $e^{0.336} \approx 1.40$ (a 40% increase, $p < 0.05$).
- **Categorical Variables:**
  - **sex:** Being male (`sexMale`) increases the odds of having heart disease by a factor of $e^{0.852} \approx 2.34$ (134% higher) compared to being female ($p < 0.01$).
  - **cp (Chest Pain Type):** Compared to being asymptomatic, all other types of chest pain are associated with significantly lower odds of disease. For instance, having *atypical angina* decreases the odds by approximately 86% ($e^{-1.976} \approx 0.14$, $p < 0.001$).
  - **exang (Exercise-Induced Angina):** Experiencing angina during exercise (`exangTRUE`) increases the odds of disease by a factor of $e^{0.918} \approx 2.50$ (150% higher) compared to not experiencing it ($p < 0.001$).

- **ca (Number of Major Vessels):** The number of blocked vessels is a powerful predictor. Having one blocked vessel (`ca1`) increases the odds by a factor of $e^{3.22} \approx 25$ compared to having none ($p < 0.001$). This effect is even stronger for two vessels.
- **thal (Thallium Scan Result):** A result of *reversible defect* (`thalreversable defect`) increases the odds of disease by a factor of $e^{0.843} \approx 2.32$ compared to the baseline category ($p < 0.05$).

## 3.7 Ordinal Logistic Regression for Disease Severity

Having established the factors for the initial diagnosis, the second stage of the analysis focuses exclusively on the subset of patients with confirmed heart disease (`num` $> 0$). The objective is to identify which clinical factors contribute to an increase in the severity of the disease, which is treated as an ordered categorical variable with four levels (1, 2, 3, 4). A Cumulative Odds Model (also known as a proportional odds logistic regression) was employed for this purpose.

### 3.7.1 Model Development and Selection

An initial full model incorporating all predictors was constructed. Subsequently, stepwise model selection using both AIC and BIC was performed to derive a more parsimonious model. The model selected by the AIC criterion was chosen for the final interpretation, as it retained a richer set of clinically relevant variables. The final model included the predictors: ‘age‘, ‘sex‘, ‘dataset‘, ‘restecg‘, ‘thalch‘, ‘oldpeak‘, and ‘ca‘.

The predictive accuracy of this model on the test set was **50.9%**. While this accuracy appears modest, it is important to note that predicting one of four ordered stages is a significantly more challenging task than binary classification. The confusion matrix revealed that the model was most effective at identifying patients in Stage 1, but struggled to accurately differentiate between the more severe stages (2, 3, and 4), which had fewer samples.

The final selected ordinal logistic regression model is specified by the following set of equations for the cumulative log-odds. Let $P(Y \leq j)$ be the cumulative probability of a patient's disease severity being at stage $j$ or less, where $j \in \{1, 2, 3\}$. The model is defined as:

$$\begin{aligned}
\text{logit}\left(P(Y \leq 1)\right) &= \log\left(\frac{P(Y \leq 1)}{1 - P(Y \leq 1)}\right) = \zeta_1 - \mathbf{X}\boldsymbol{\beta} \\
\text{logit}\left(P(Y \leq 2)\right) &= \log\left(\frac{P(Y \leq 2)}{1 - P(Y \leq 2)}\right) = \zeta_2 - \mathbf{X}\boldsymbol{\beta} \\
\text{logit}\left(P(Y \leq 3)\right) &= \log\left(\frac{P(Y \leq 3)}{1 - P(Y \leq 3)}\right) = \zeta_3 - \mathbf{X}\boldsymbol{\beta}
\end{aligned}$$

where the linear component $\mathbf{X}\boldsymbol{\beta}$ is given by:

$$\begin{aligned}
\mathbf{X}\boldsymbol{\beta} = &- 0.0339 \cdot \text{age} + 0.5286 \cdot \text{sex}_{\text{Male}} \\
&- 18.3186 \cdot \text{dataset}_{\text{Hungary}} - 0.2063 \cdot \text{dataset}_{\text{Switzerland}} - 0.4200 \cdot \text{dataset}_{\text{VA Long Beach}} \\
&- 0.2060 \cdot \text{restecg}_{\text{lv hypertrophy}} - 0.3815 \cdot \text{restecg}_{\text{normal}} + 0.2774 \cdot \text{restecg}_{\text{st-t abnormality}} \\
&- 0.0189 \cdot \text{thalch} + 0.5449 \cdot \text{oldpeak} \\
&+ 0.3767 \cdot \text{ca}_1 + 1.5338 \cdot \text{ca}_2 + 1.6545 \cdot \text{ca}_3
\end{aligned}$$

The intercepts (cut-points), denoted by $\zeta_j$, define the thresholds between the cumulative probabilities and are estimated as:

- $\zeta_1$ (Threshold 1|2) = -3.3959
- $\zeta_2$ (Threshold 2|3) = -1.9869
- $\zeta_3$ (Threshold 3|4) = 0.1760

### 3.7.2 Interpretation of Coefficients

The coefficients from the ordinal logistic regression model represent the change in the log-odds of a patient being at a higher level of disease severity versus a lower level. For interpretation, we exponentiate the coefficients to obtain the proportional odds ratio. An odds ratio greater than 1 indicates that an increase in the predictor is associated with an increased likelihood of the disease being more severe.

- **age:** For each additional year of age, the odds of being in a more severe disease category (e.g., Stage 2 or higher vs. Stage 1) decrease by approximately 3.3% ($e^{-0.03385} \approx 0.967$). This suggests that among patients who already have the disease, younger age is paradoxically associated with a more severe diagnosis in this dataset.
- **sex:** Being male (`sexMale`) increases the odds of being in a more severe disease category by a factor of $e^{0.5286} \approx 1.70$. This means males have 70% higher odds of having a more advanced stage of heart disease compared to females, holding other factors constant.
- **thalch (Max Heart Rate Achieved):** A higher maximum heart rate is associated with a less severe form of the disease. For each one-beat-per-minute increase in ‘thalch‘, the odds of being in a higher severity category decrease by about 1.9% ($e^{-0.01887} \approx 0.981$).
- **oldpeak (ST Depression):** This is a strong indicator of disease severity. For each one-unit increase in ST depression, the odds of the disease being more severe increase by a factor of $e^{0.5449} \approx 1.72$. This translates to a 72% increase in the odds of advancing to a higher disease stage.
- **ca (Number of Major Vessels):** The number of blocked vessels is a critical factor in disease progression. Compared to having no blocked vessels, having two blocked vessels (`ca2`) increases the odds of being in a more severe category by a factor of $e^{1.5338} \approx 4.64$ (a 364% increase). The effect is even stronger for three blocked vessels (`ca3`), which increases the odds by a factor of $e^{1.6545} \approx 5.23$ (a 423% increase).
- **restecg (Resting ECG Results):** The coefficients for ‘restecg‘ were not statistically significant in the final model, suggesting that while it may be relevant for initial diagnosis, it does not strongly differentiate between the stages of disease severity in this patient cohort.

In summary, among patients with existing heart disease, factors such as higher ST depression (‘oldpeak‘) and a greater number of blocked vessels (‘ca‘) are the strongest drivers of disease severity. Conversely, a higher maximum heart rate (‘thalch‘) is associated with a less advanced stage of the disease.

## 3.8 Model Evaluation

To assess the practical performance of the final selected models, a comprehensive evaluation was conducted on the independent testing set. Different metrics were used for the binary and ordinal models to reflect their distinct objectives.

### 3.8.1 Binary Classification Model Evaluation

The performance of the final binary logistic regression model (selected via BIC) was evaluated based on its ability to accurately classify patients as either having heart disease or not.

#### 3.8.1.1 ROC and AUC Analysis:

The model's overall discriminatory power was assessed using the Receiver Operating Characteristic (ROC) curve. The model achieved an excellent **Area Under the Curve (AUC) of 0.957**, indicating a very high capability to distinguish between patients with and without heart disease across all probability thresholds.

#### 3.8.1.2 Confusion Matrix and Performance Metrics:

Using a standard probability threshold of 0.5, the model's performance was further detailed in a confusion matrix. The key metrics derived are presented in Table 8.

Table 8: Performance Metrics for the Final Binary Classification Model

| Metric | Value |
|---|---|
| Accuracy | 88.04% |
| 95% Confidence Interval | (79.6%, 93.9%) |
| Sensitivity (Recall) | 87.27% |
| Specificity | 89.19% |
| Positive Predictive Value (Precision) | 92.31% |
| Negative Predictive Value | 82.50% |
| F1-Score | 0.8974 |
| Kappa | 0.7546 |

The model demonstrates strong and balanced performance, with an overall **accuracy of 88.0%**. The high **Sensitivity (87.3%)** indicates that the model correctly identifies 87.3% of all patients who do not have heart disease. Similarly, the high **Specificity (89.2%)** shows that it correctly identifies 89.2% of patients who do have heart disease[1]. A **Kappa value of 0.755** signifies a substantial level of agreement between the predictions and the actual outcomes, correcting for chance.

### 3.8.2 Ordinal Classification Model Evaluation

Evaluating the ordinal model is inherently more challenging, as its goal is to predict one of four ordered stages of disease severity.

#### 3.8.2.1 Confusion Matrix and Accuracy:

The final ordinal model (selected via AIC) achieved an overall **accuracy of 50.9%** on the test set of diseased patients. As shown in the confusion matrix, the model performed best at identifying patients in Stage 1, correctly classifying 23 out of 28 cases (a sensitivity of 82.1%). However, its ability to distinguish between the more severe and less populated stages (2, 3, and 4) was significantly weaker, with sensitivity dropping to 9.1% for Stage 2 and 28.6% for Stage 3. No patients in Stage 4 were correctly identified.

[1]Note: The R output's Sensitivity/Specificity labels depend on the 'positive' class definition. Based on the confusion matrix, we have re-labeled them for clarity where 'diseased=1' is the condition of interest.

#### 3.8.2.2 Kappa Statistic:

To account for agreement by chance, Cohen's Kappa was calculated. The model yielded a **Kappa value of 0.154**, indicating only a slight level of agreement. This result underscores the difficulty of the multi-class ordinal prediction task, especially with an imbalanced distribution of severity stages in the test set.

### 3.8.3 Overall Conclusion

The binary logistic regression model proved to be highly effective and reliable for the primary task of diagnosing the presence of heart disease, achieving high accuracy and an excellent AUC. In contrast, while the ordinal model successfully identified several key factors related to disease progression, its practical predictive accuracy for specific severity stages is limited. This suggests that while it provides valuable insights into risk escalation, its use as a precise multi-stage diagnostic tool would require further refinement or a larger, more balanced dataset.